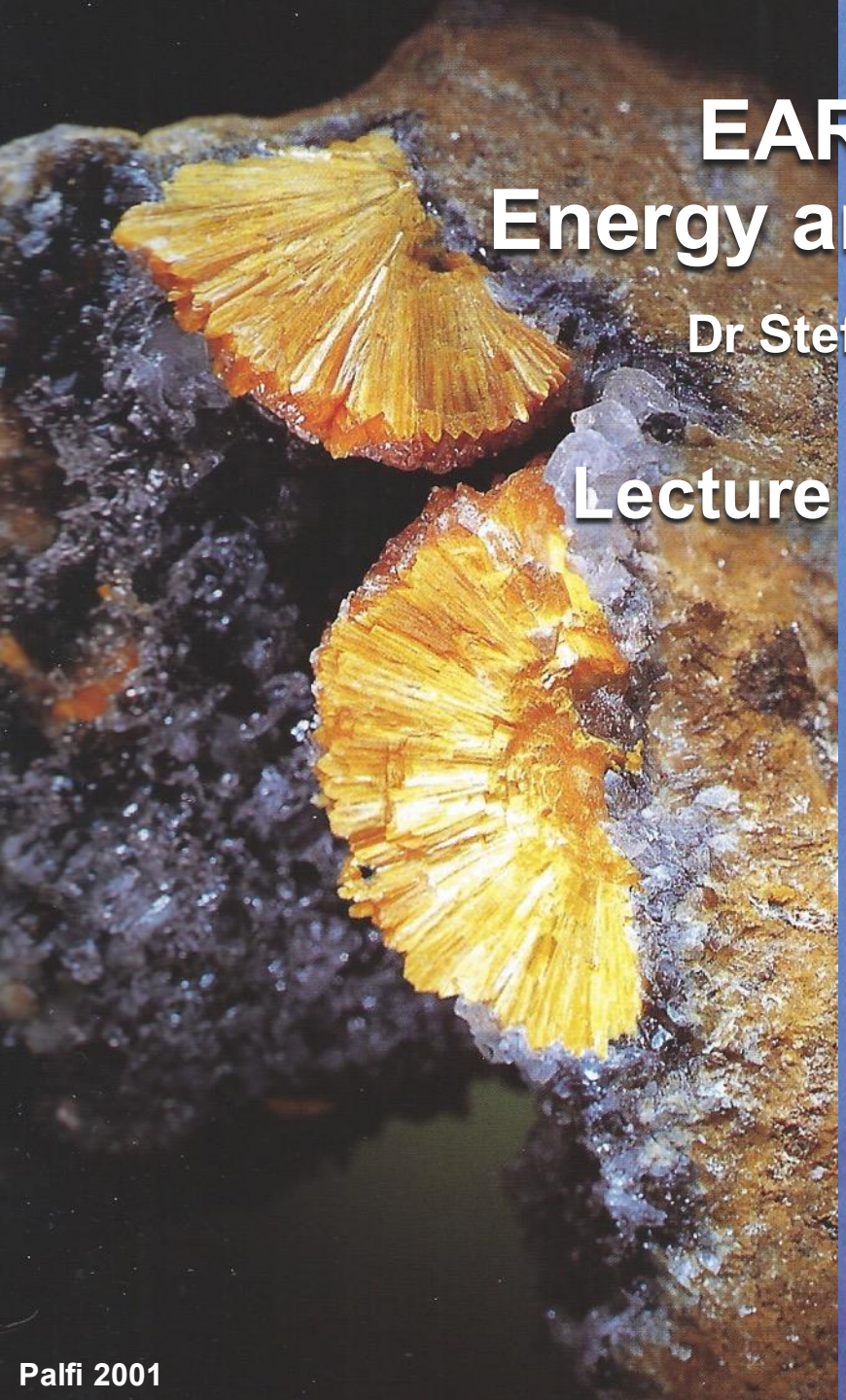
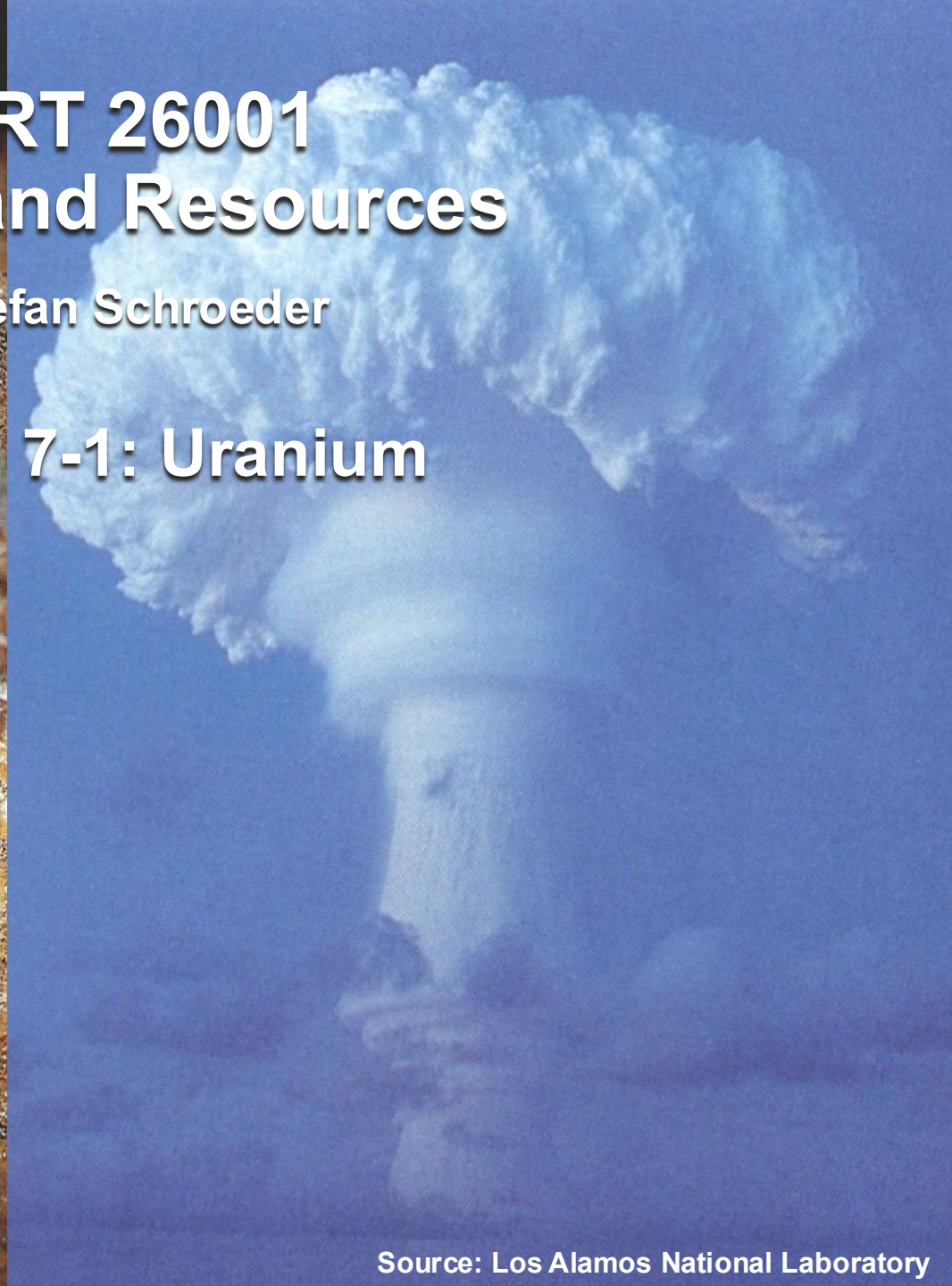
# EART 26001
# Energy and Resources

Dr Stefan Schroeder

## Lecture 7-1: Uranium





Palfi 2001

Source: Los Alamos National Laboratory

# Energy and Resources
## Learning goals week 7

**This week you will learn to:**

- **Describe the uses, physical and chemical properties of uranium.**
- **Recognize the hydrothermal processes forming uranium deposits.**
- **Describe the fundamental physics of nuclear chain reaction, and the operation of nuclear power generation, via the natural reactor in Gabon.**
- **Recognize the exploration, production methods, and environmental impact of uranium mining.**

# Uranium properties + applications
## Radioactive decay and nuclear fission




Press&Siever 1985

- **Application in radiometric dating:**

$$N = N_0 \times e^{\lambda t}$$

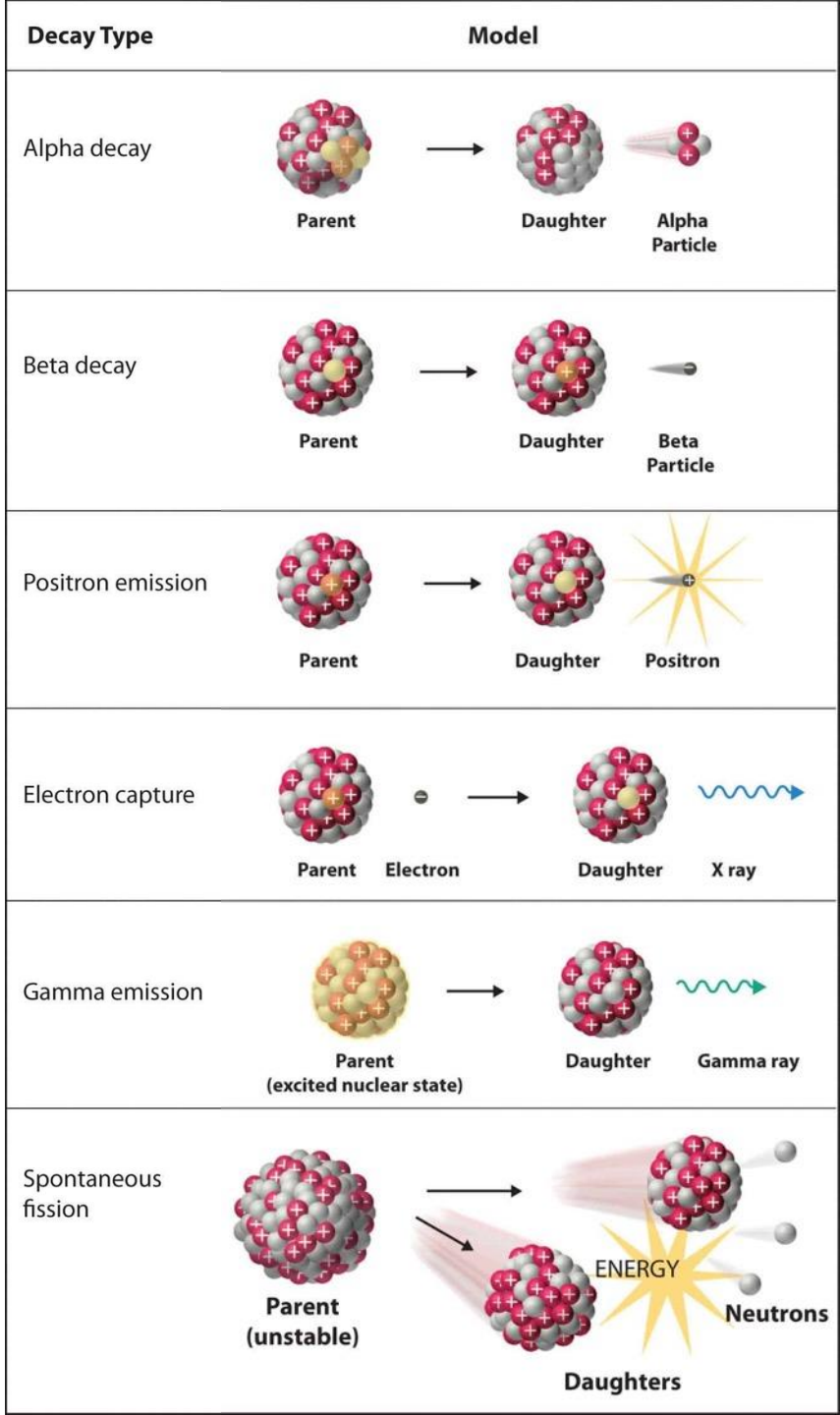
**N: measured radioactive nuclides**

**$N_0$: original radioactive nuclides**

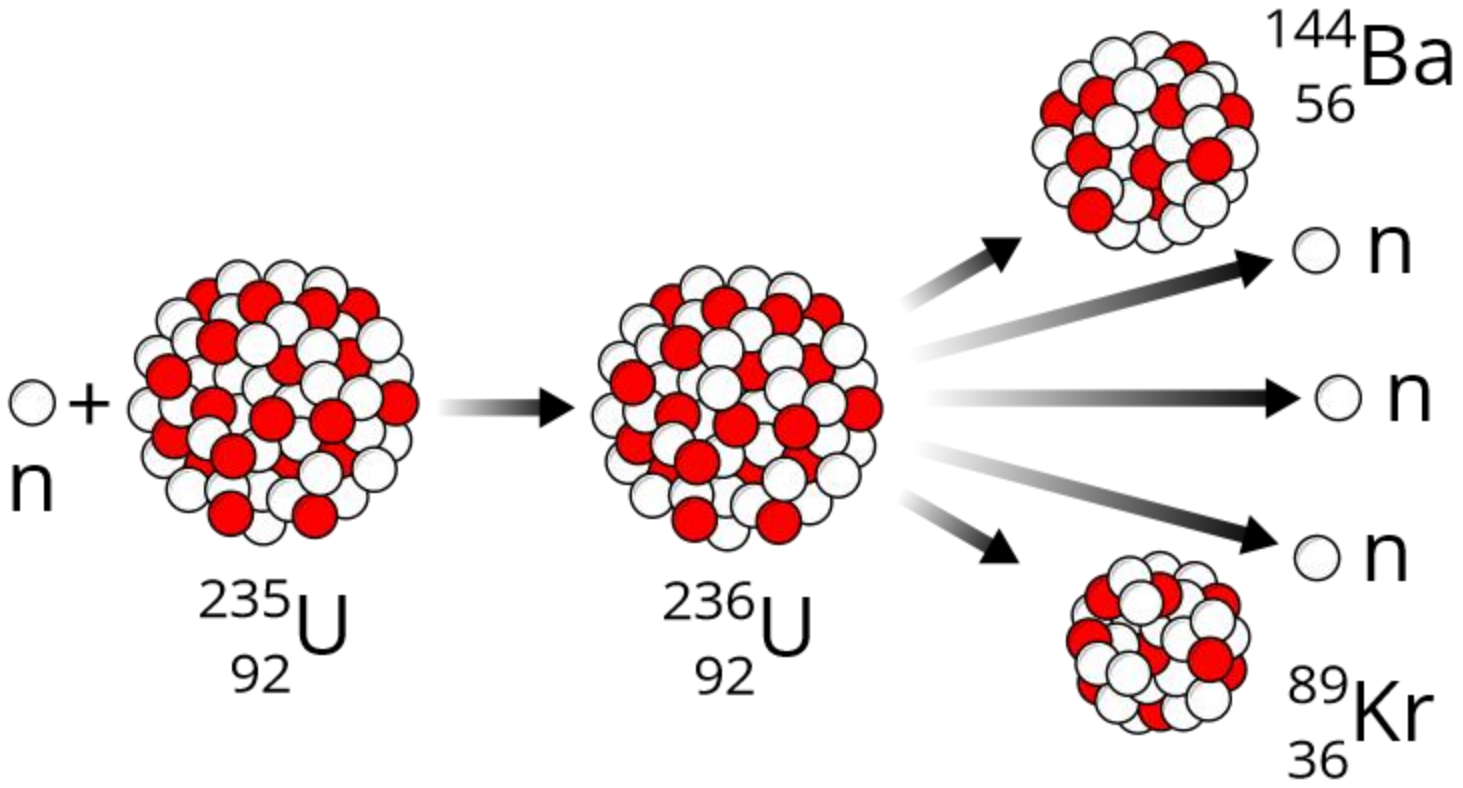
**$\lambda$: decay constant**

**t: time units**





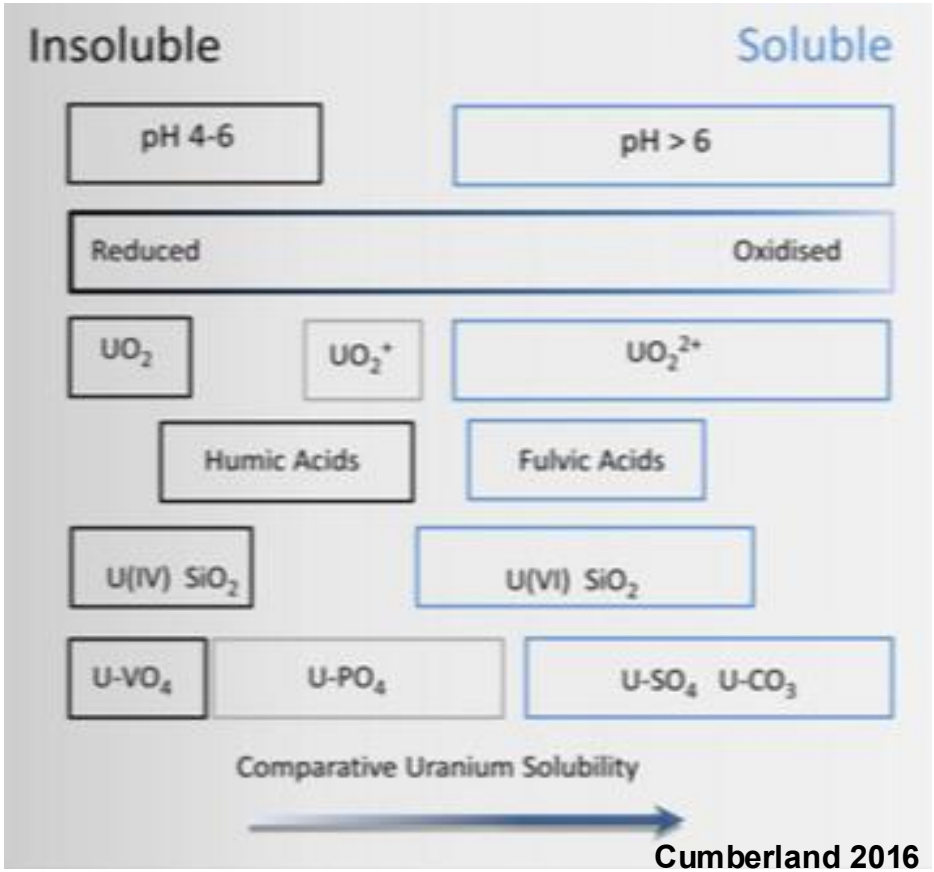
# Uranium properties and applications
## Radioactive decay and nuclear fission



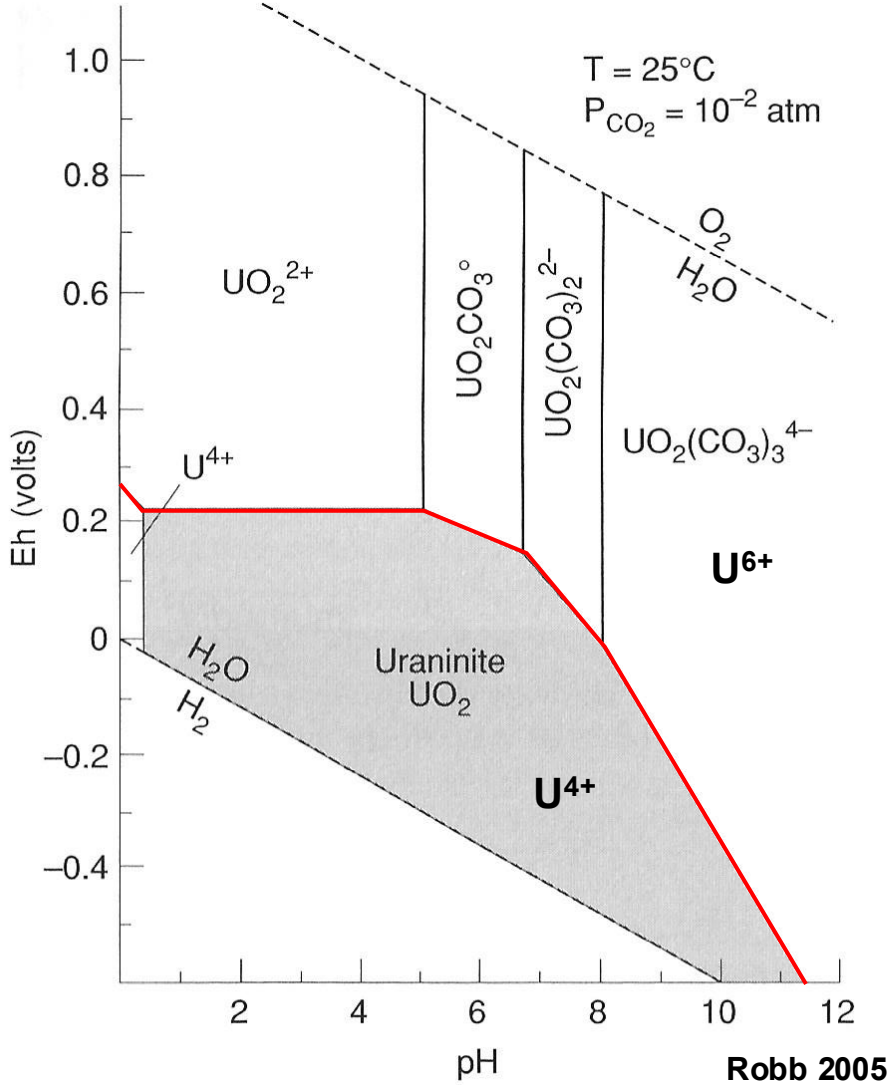

© Wikipedia

- Application in controlled (nuclear reactor) and uncontrolled (atomic bomb) chain reactions

# Uranium properties
## Redox geochemistry




Cumberland 2016




Robb 2005

- Uranous $U^{4+}$ stable in reducing conditions, relatively insoluble
- Uranic $U^{6+}$ stable in oxidizing conditions, relatively soluble, forms stable aqueous complexes ($CO_3^{2-}$, $OH^-$, $H^-$, $PO_4^{2-}$, $F^-$, $Cl^-$), and minerals when cations are available

# Uranium
## Minerals

- Uraninite $U^{IV}O_2$
- In nature partly oxidized as pitchblende: $U^{IV}{}_3O_8$
- Brannerite $(U^{IV},Ca,Ce)(Ti,Fe)_2O_6$

- Carnotite $K_2(U^{VI}O_2)_2(VO_4)_2 \cdot 3H_2O$
- Autunite $Ca(U^{VI}O_2)_2(PO_4)_2 \cdot 10H_2O$
- etc

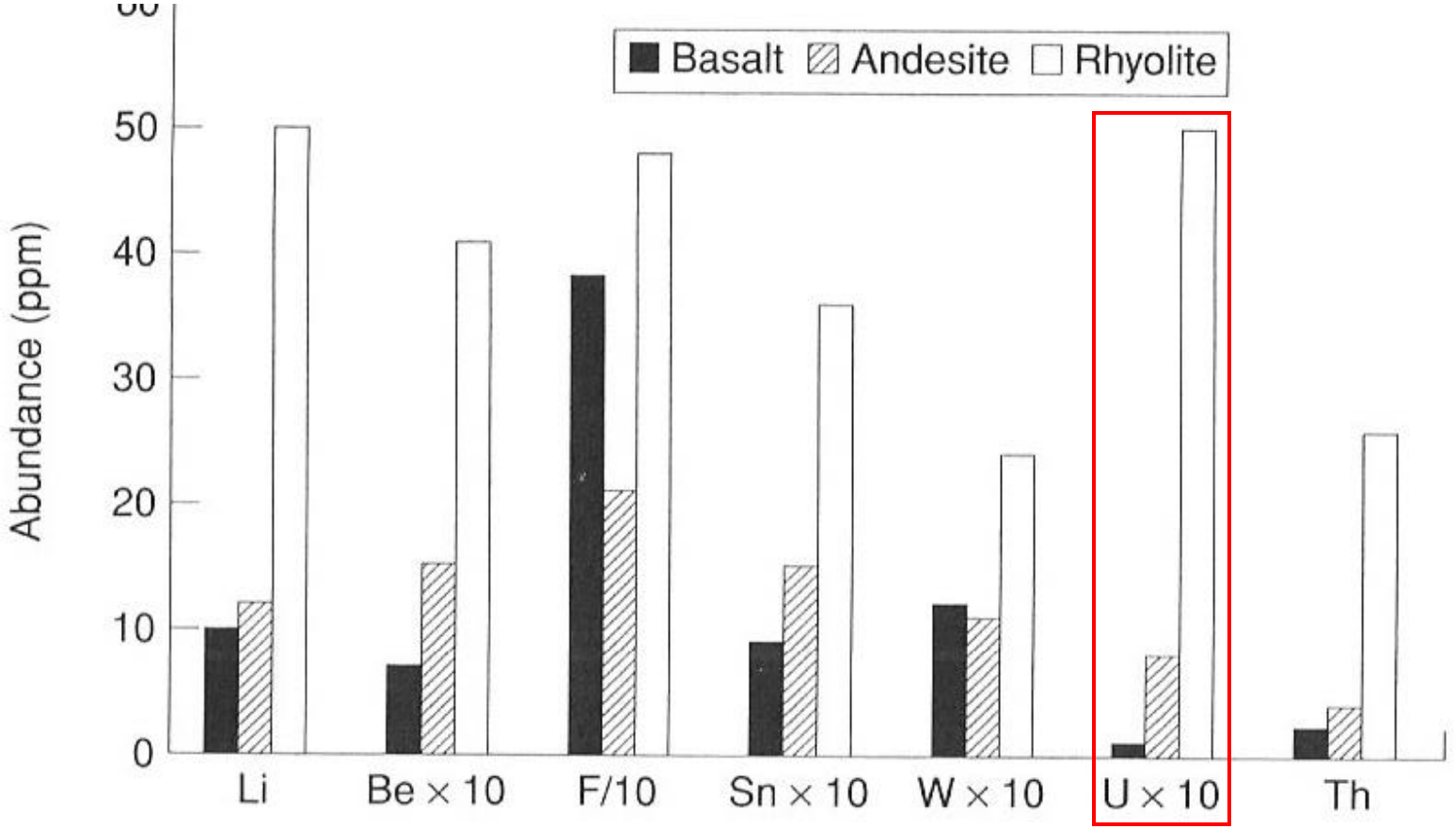








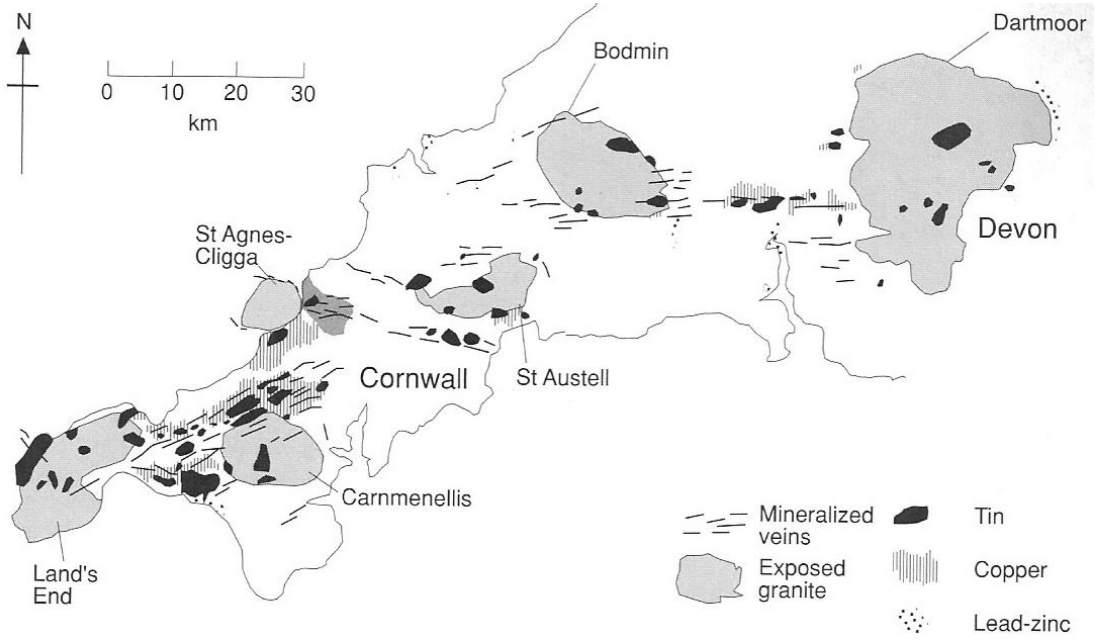
# Uranium properties and applications
## Lithophile behavior



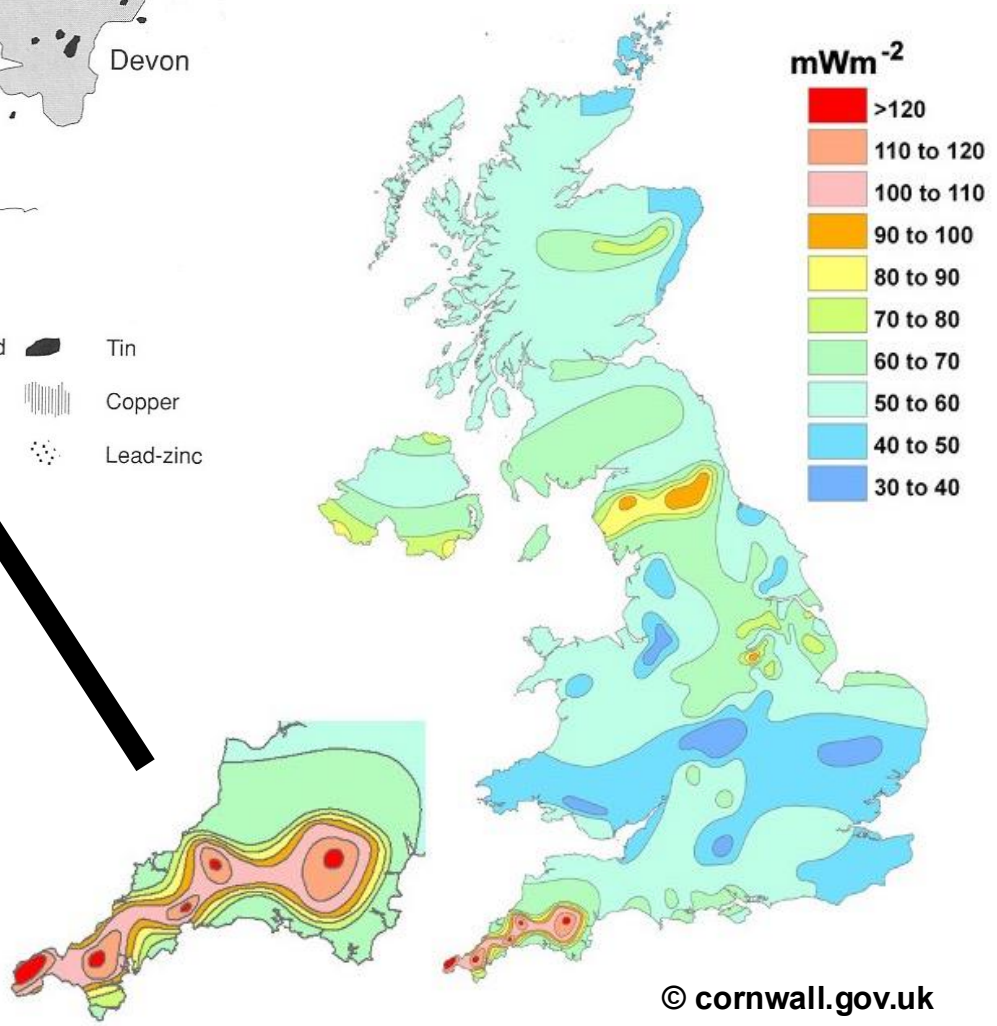

Robb 2008

- **Enriched in granitic lithologies and their late-stage crystallization products (e.g. zircon, monazite)**

# Uranium properties and applications
## Lithophile behavior




Robb 2008

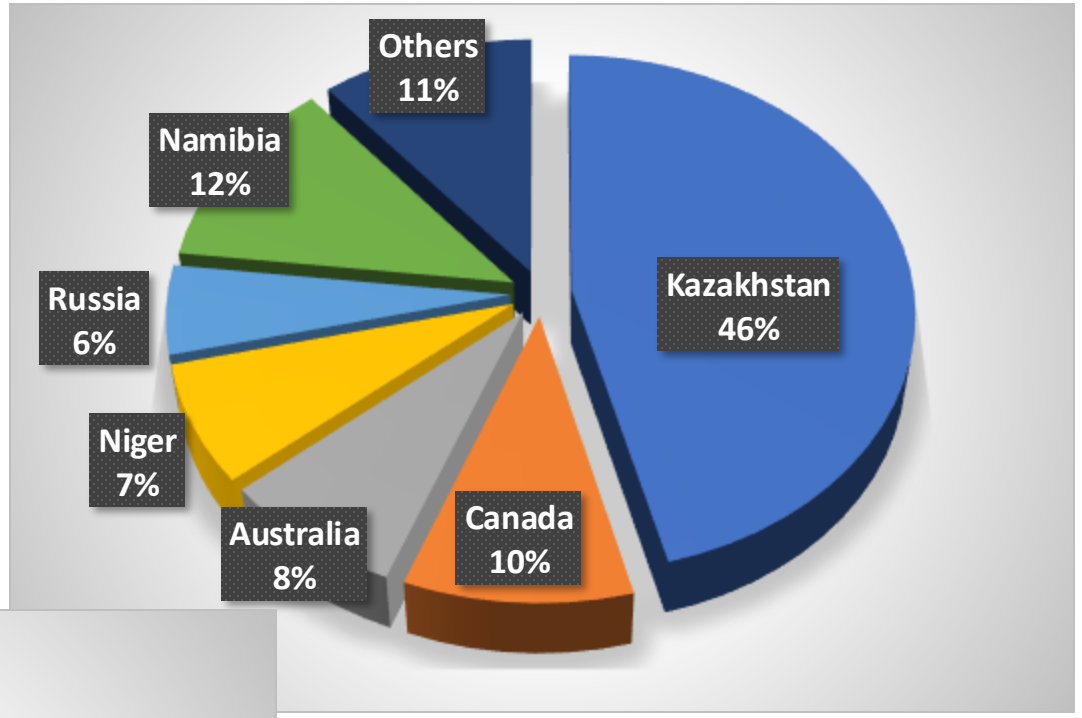



- U/Th-bearing granites: high heat producing (HHP) granites
- Magmatic and magmatic-hydrothermal mineralization
- Potential for geothermal projects

© cornwall.gov.uk

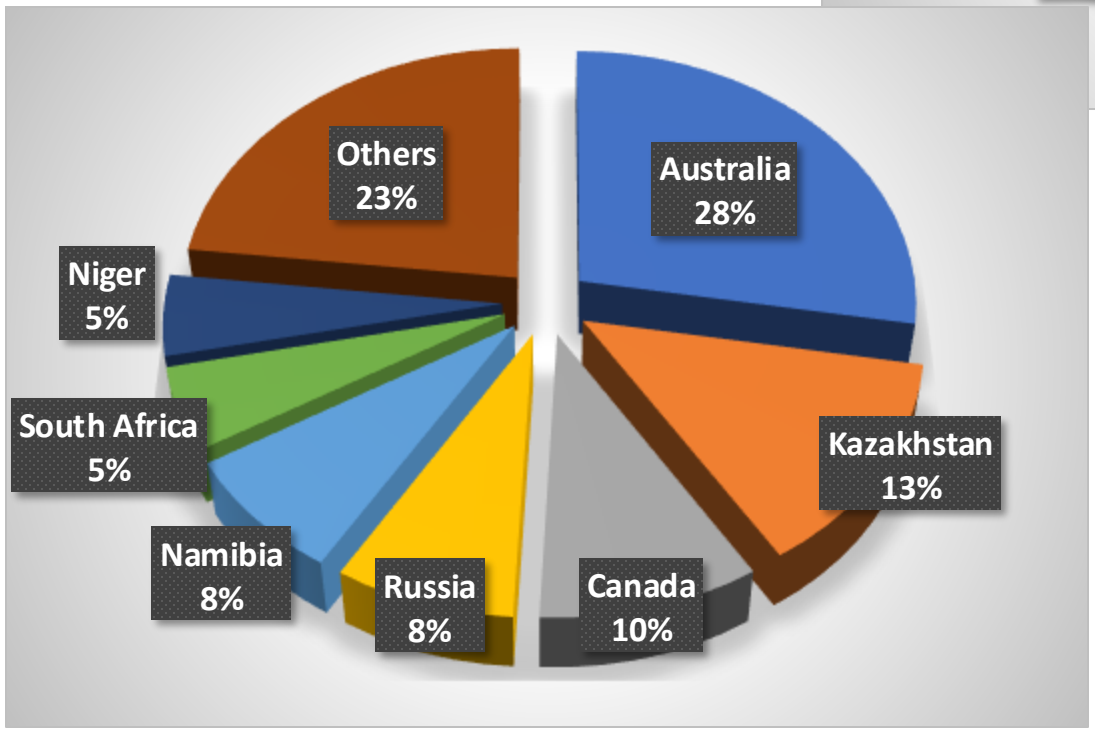
# Uranium
## Reserves

U mine production 2021




Source : IAEA

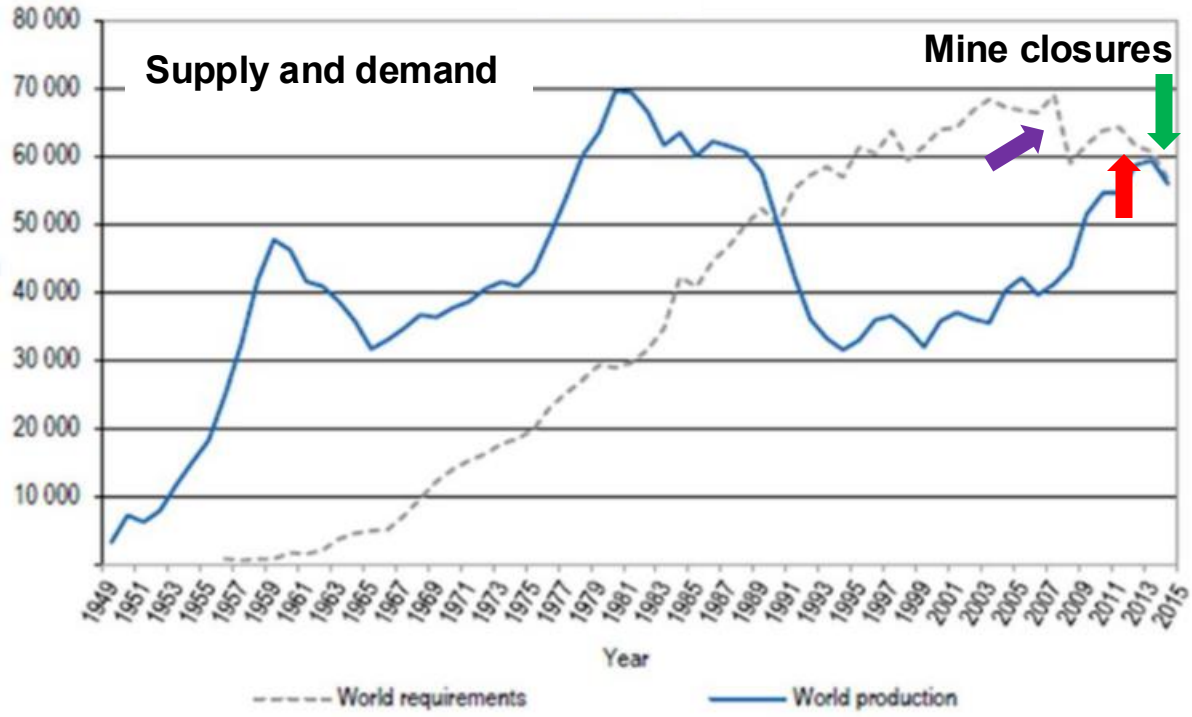
U Identified Recoverable Resources 2021*



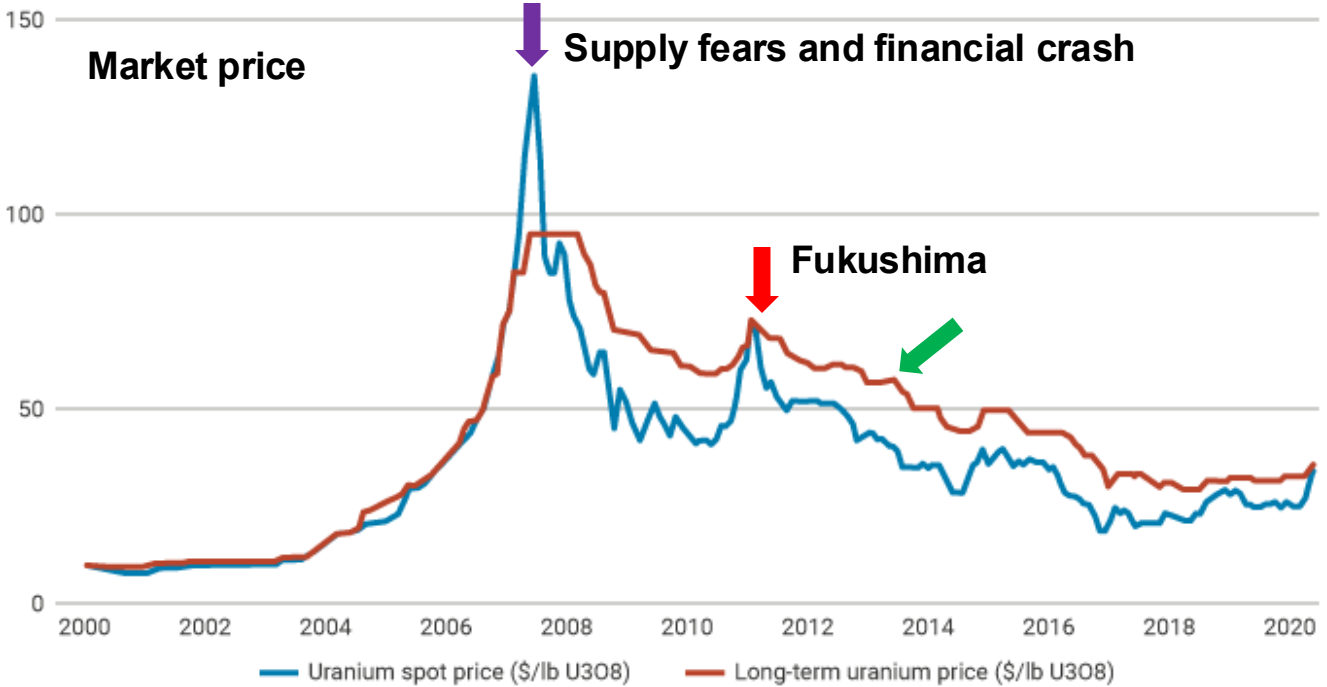

*Market price <130$/kg U

Source : IAEA

# Uranium
## Market




Source : IAEA




Source : world-nuclear.org

# EART 26001
# Energy and Resources

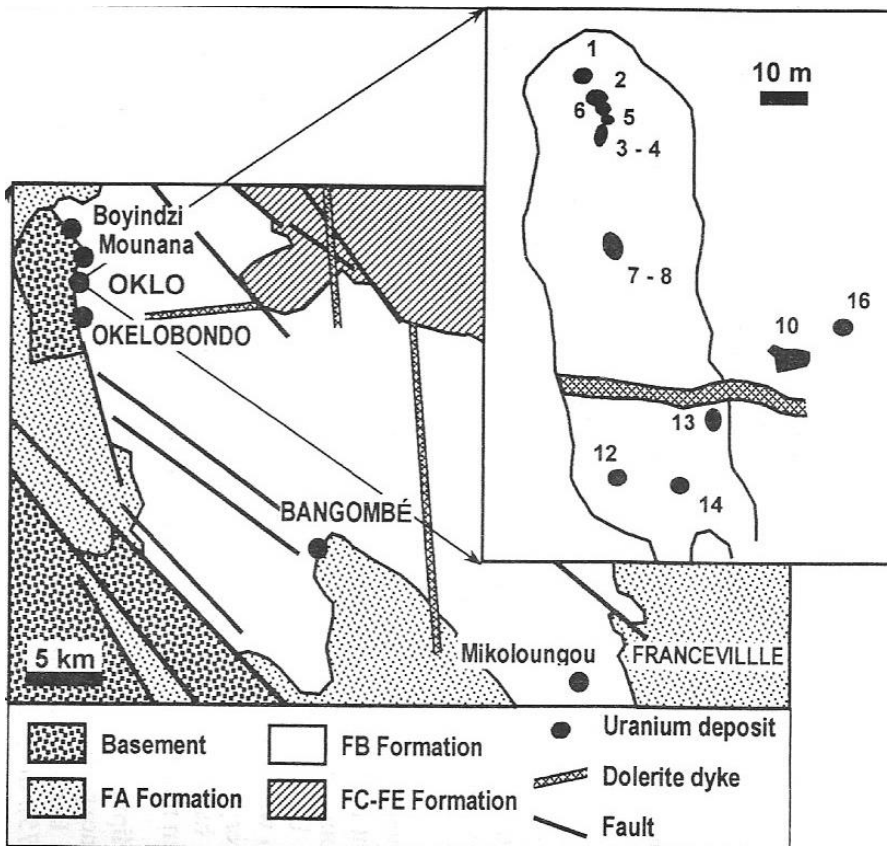
Dr Stefan Schroeder

## Lecture 7-2: The natural reactor at Oklo (Gabon)

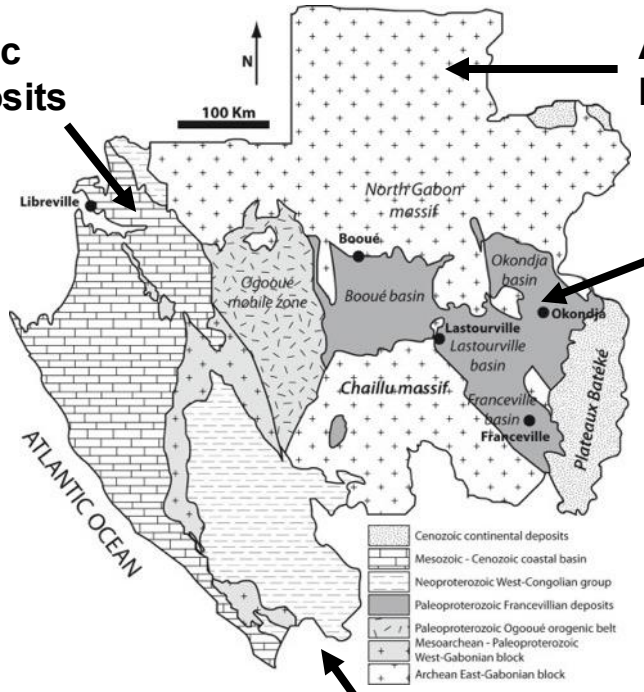


Source:
Pourlascience.fr

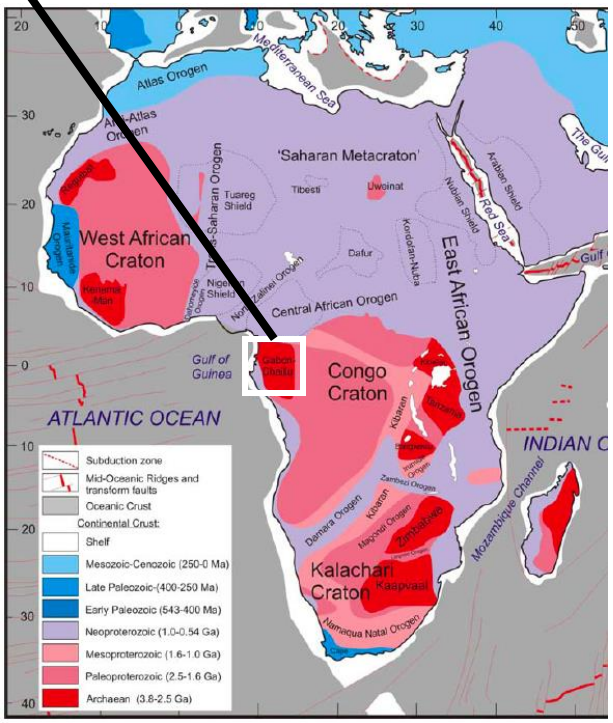
# Geological Setting
## Structure




Janeczek 1999




Lecomte 2020

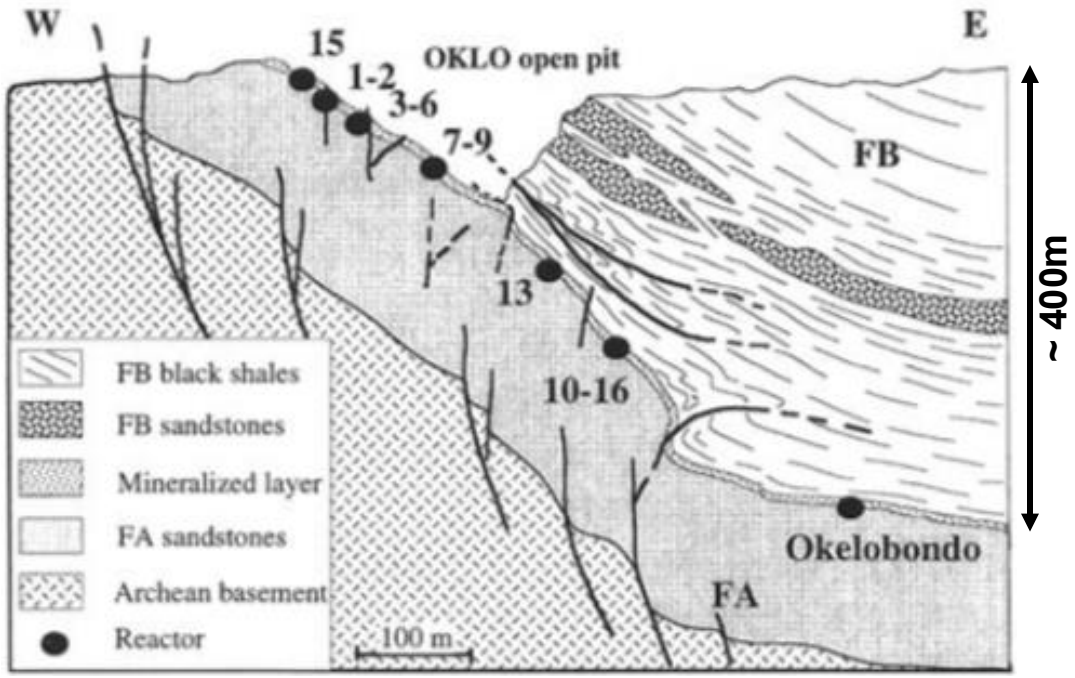



Van Hinsbergen 2011, after Gubanov & Mooney 2009

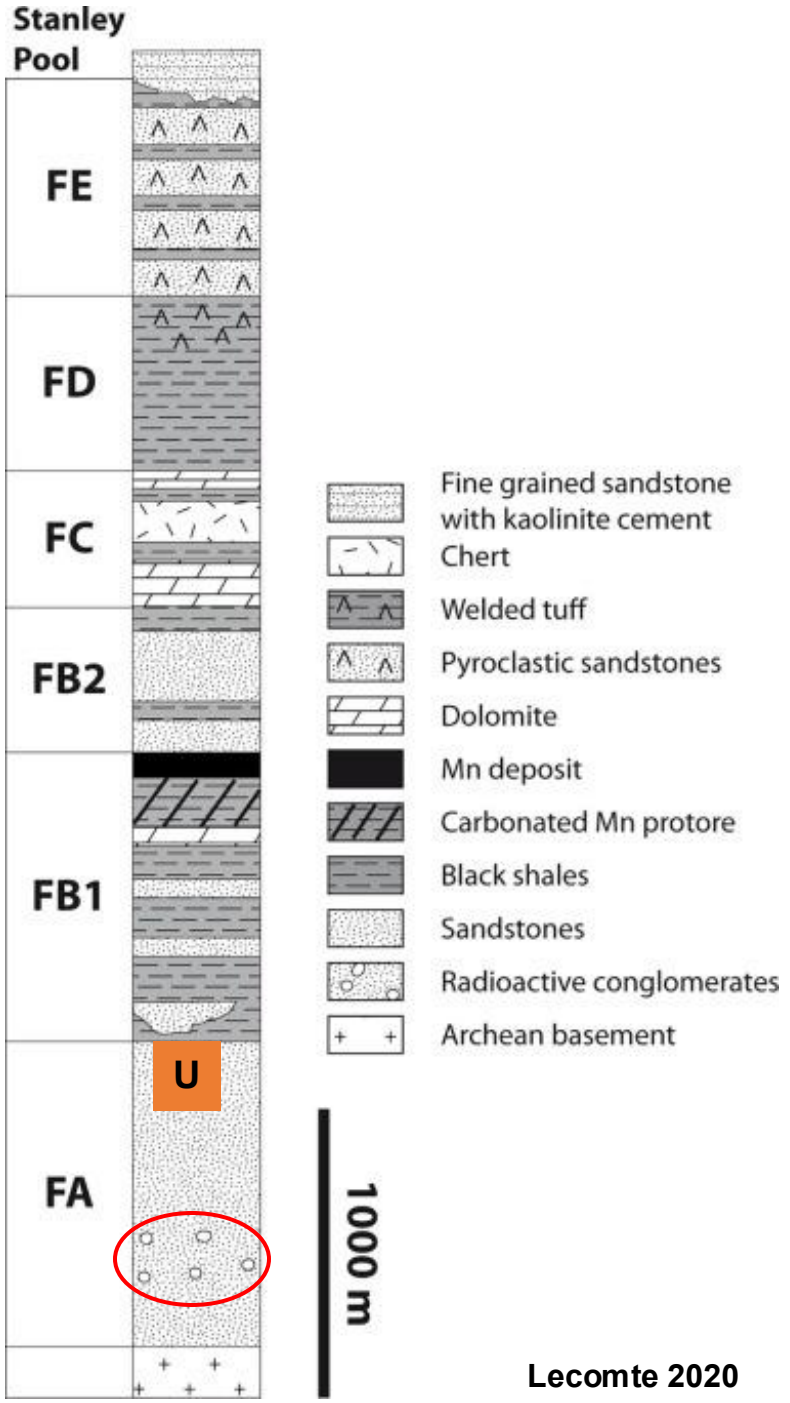
- **Proterozoic sedimentary basin (~2.1 Ga) on Archean cratonic basement**

# Geological Setting
## Stratigraphy

- Fluvio-deltaic conglomerates and sandstones (FA), organic-rich shales (FB)
- U-bearing basal conglomerates




Gauthier-Lafaye 1996



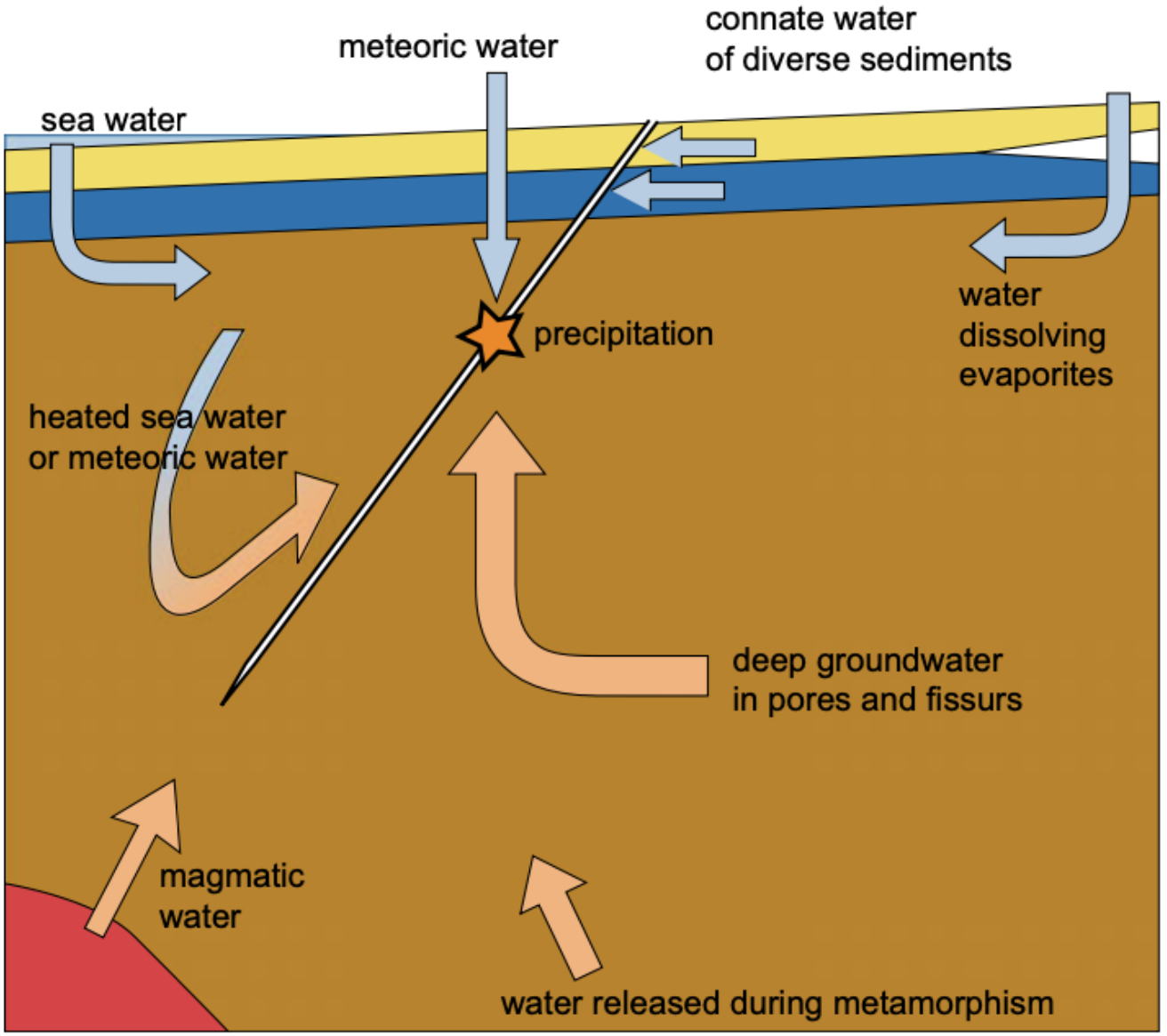

Lecomte 2020

# Uranium mineralization
## Mining



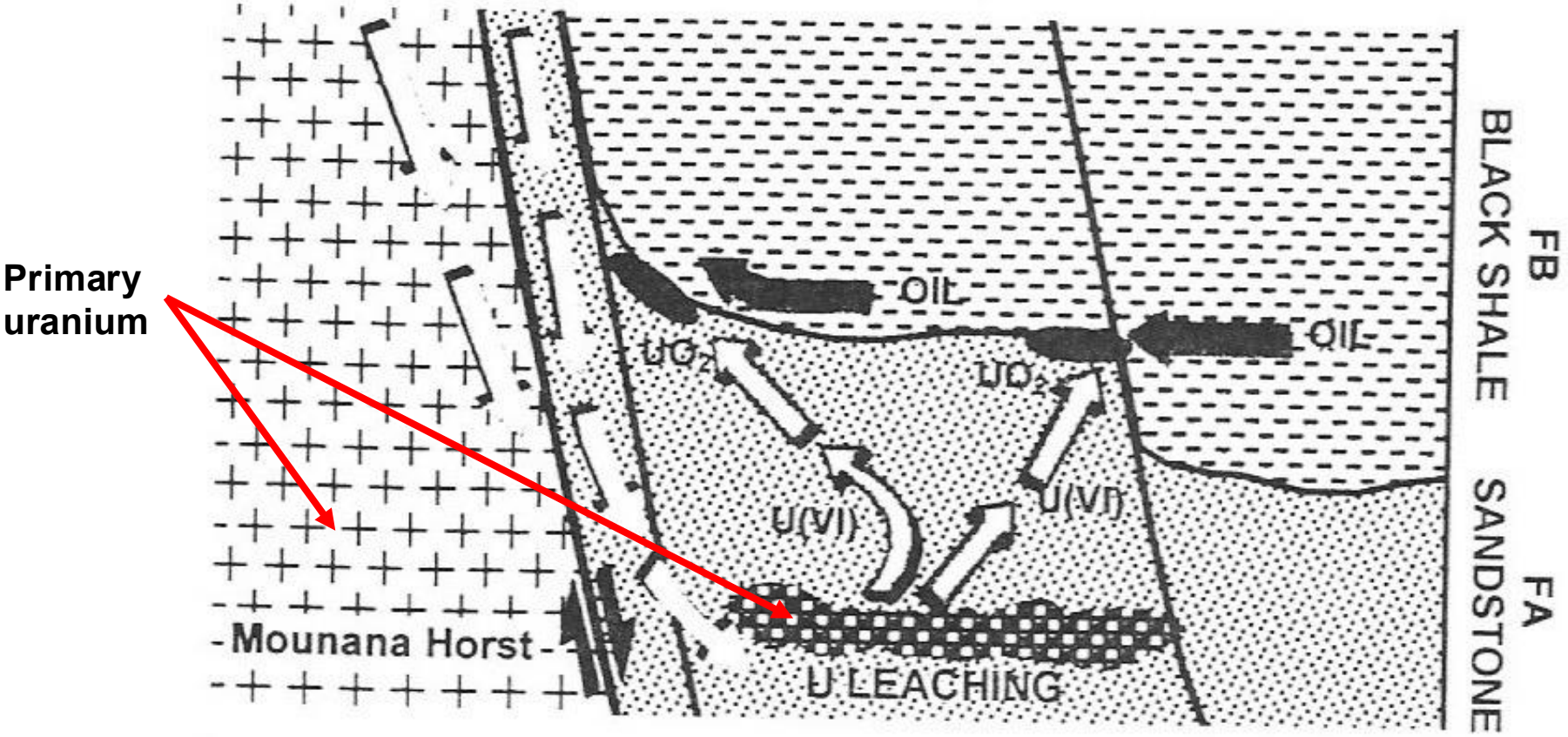
© Gauthier-Lafaye, Strasbourg

# Uranium mineralization
## Hydrothermal processes




Neukirchen 2020

# Uranium mineralization
## Hydrothermal processes



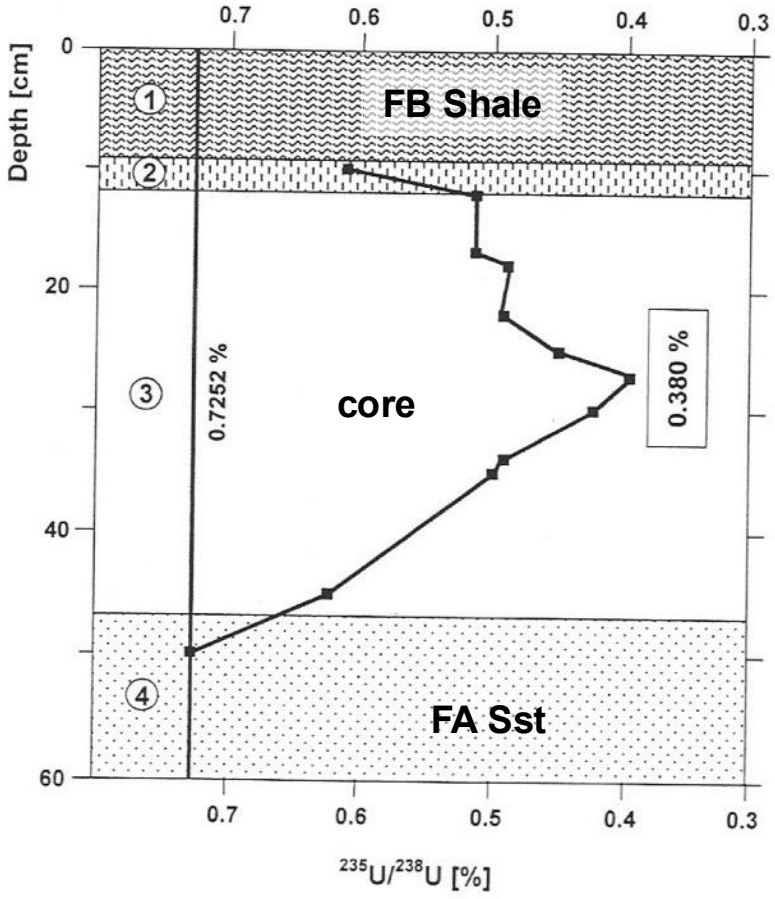

Janeczek 1999

- Hydrothermal mineralizing system at ~2.05 Ga, fluids 100-170°C
- Descending oxidizing meteoric waters
- $U^{6+}$ fluids reduced at contact with kerogen and $CH_4$ from FB shale
- Mineralization with 10-20% U

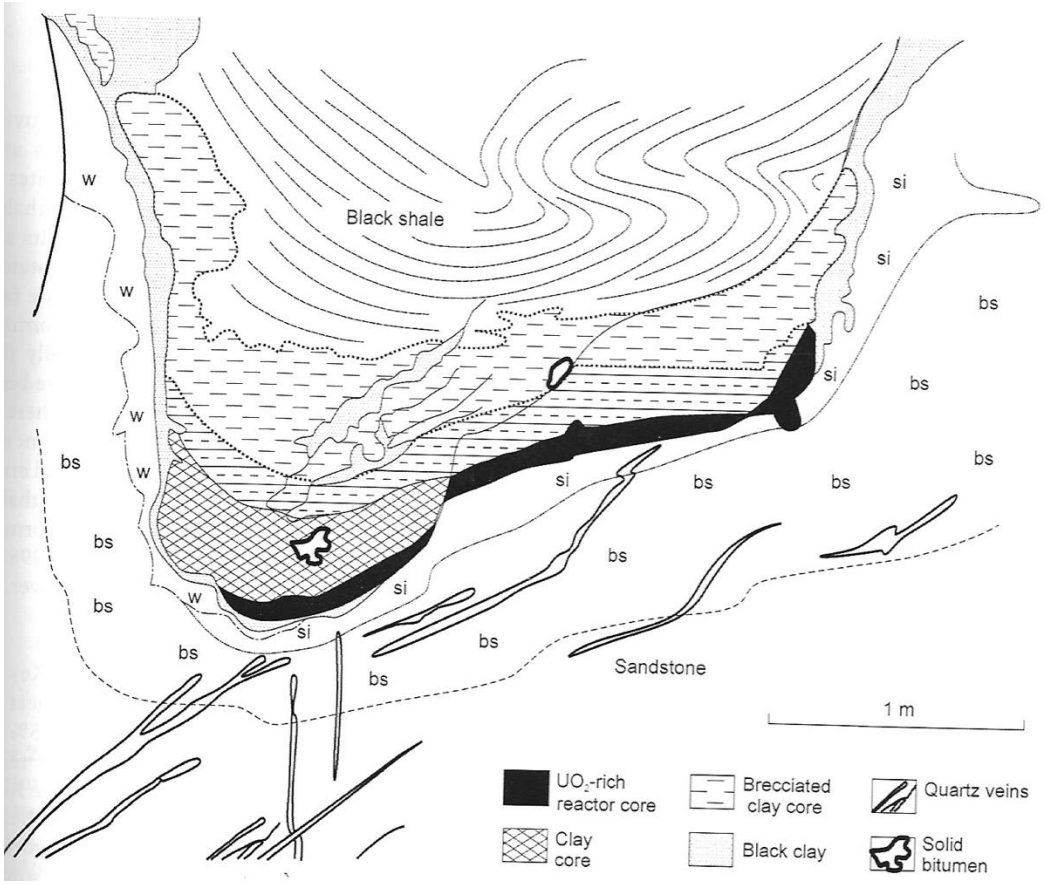
# Uranium reactor
## Reactor core alteration

- Reactor lenses <40 x 6 x 0.5 m
- m-scale synforms due to sandstone dissolution (83% loss of silica)
- $^{235}U/^{238}U$ ratio indicates chain reaction has occurred



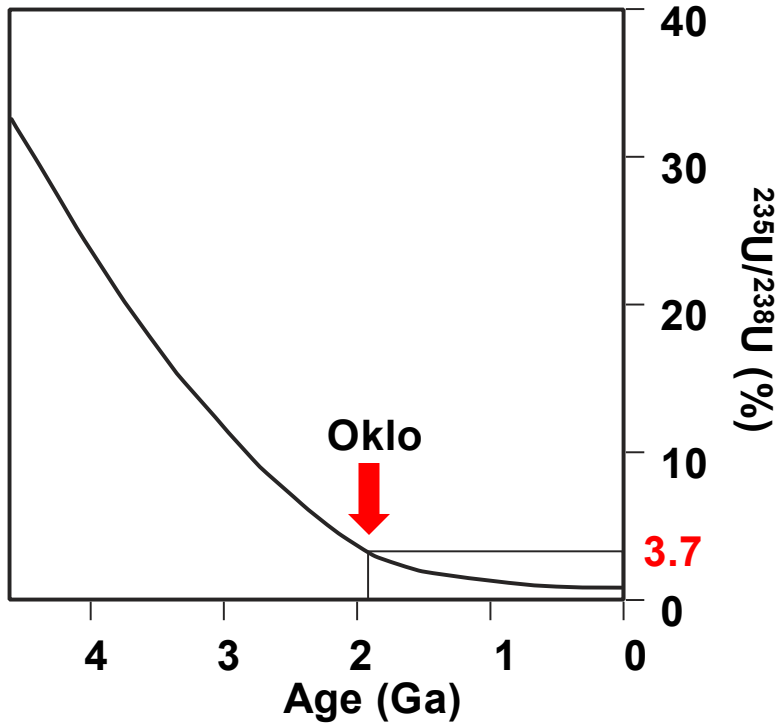

Janeczek 1999

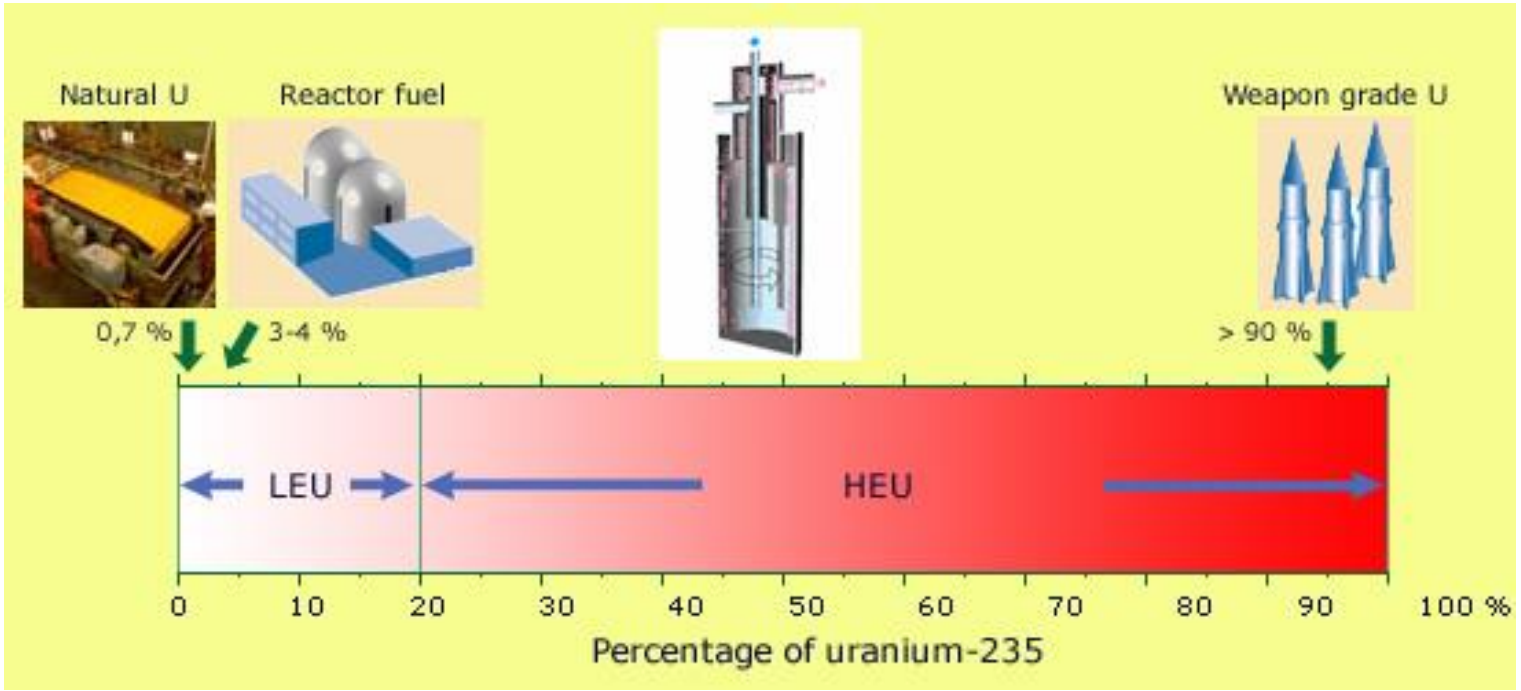



Pohl 2005

# Uranium reactor
## Conditions for reactor criticality

- $^{235}U/^{238}U$ > 1%: faster decay of $^{235}U$ over $^{238}U$ means ratio has changed with time
- >10-20% U: mineralization
- Minimum ore thickness >50 cm
- Enough water to slow fast neutrons: high water/rock ratios during mineralization
- Quartz in sandstone to reflect neutrons back into core

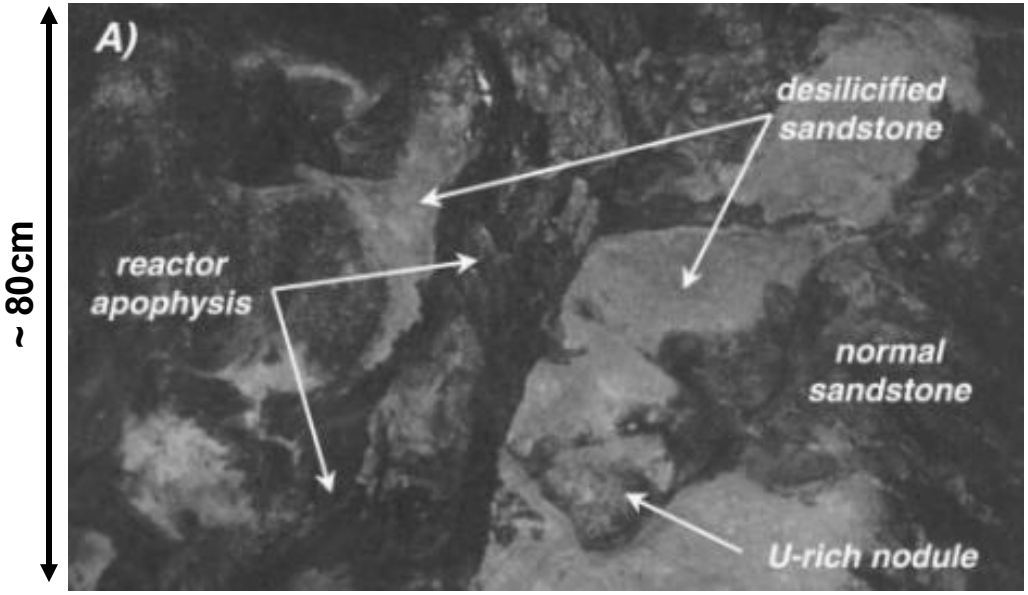


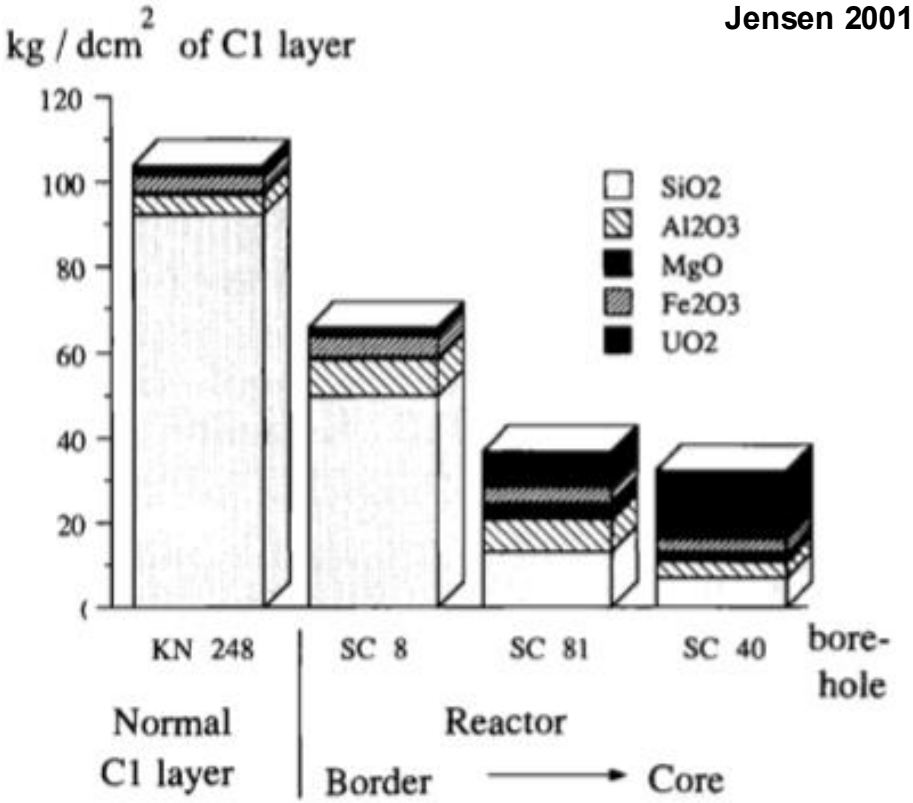

Source: www.radioactivity.eu.com

# Uranium reactor
## Chain reaction

- Duration 20,000 – 600,000 years
- 100kW/reactor ($10^{-5}$ x modern reactors)
- Temperatures:
  - margin 250-450°C
  - core <1000°C
  - gradient: 100°C/m
- Hydrothermal system: desilicified sandstone, uraninite, galena, clay minerals
- U 30-90%



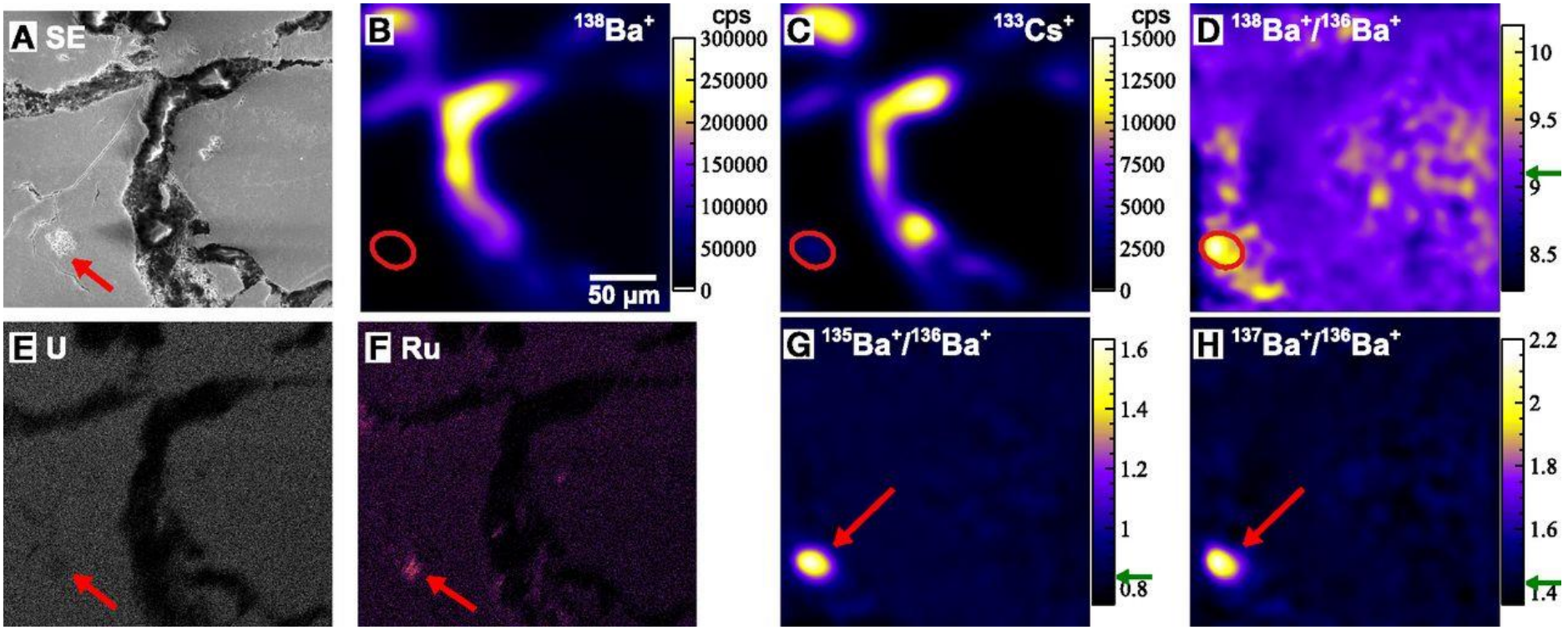

Jensen 2001




Gauthier-Lafaye 1989

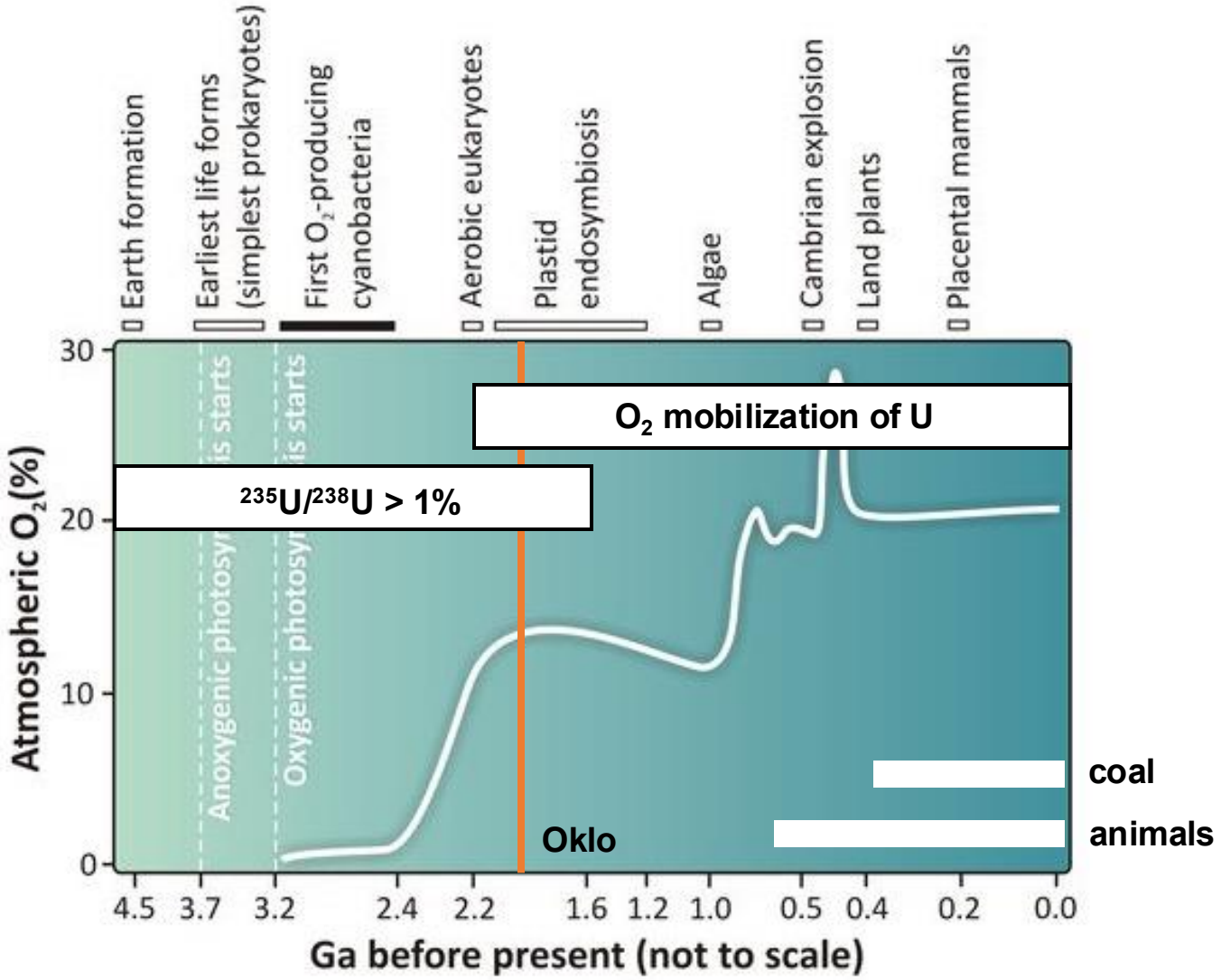
# Uranium reactor
## Radioactive waste

- Site is still mildly radioactive
- Majority of waste products have been removed
- Evidence for trapping of Ba, Cs, I, Rb, Sr, U fission products in various associated minerals (clays, phosphates, oxy-hydroxides) shortly after end of criticality




Groopman 2018

# Uranium reactor
## Implications




© NASA Astrobiology

# EART 26001
# Energy and Resources

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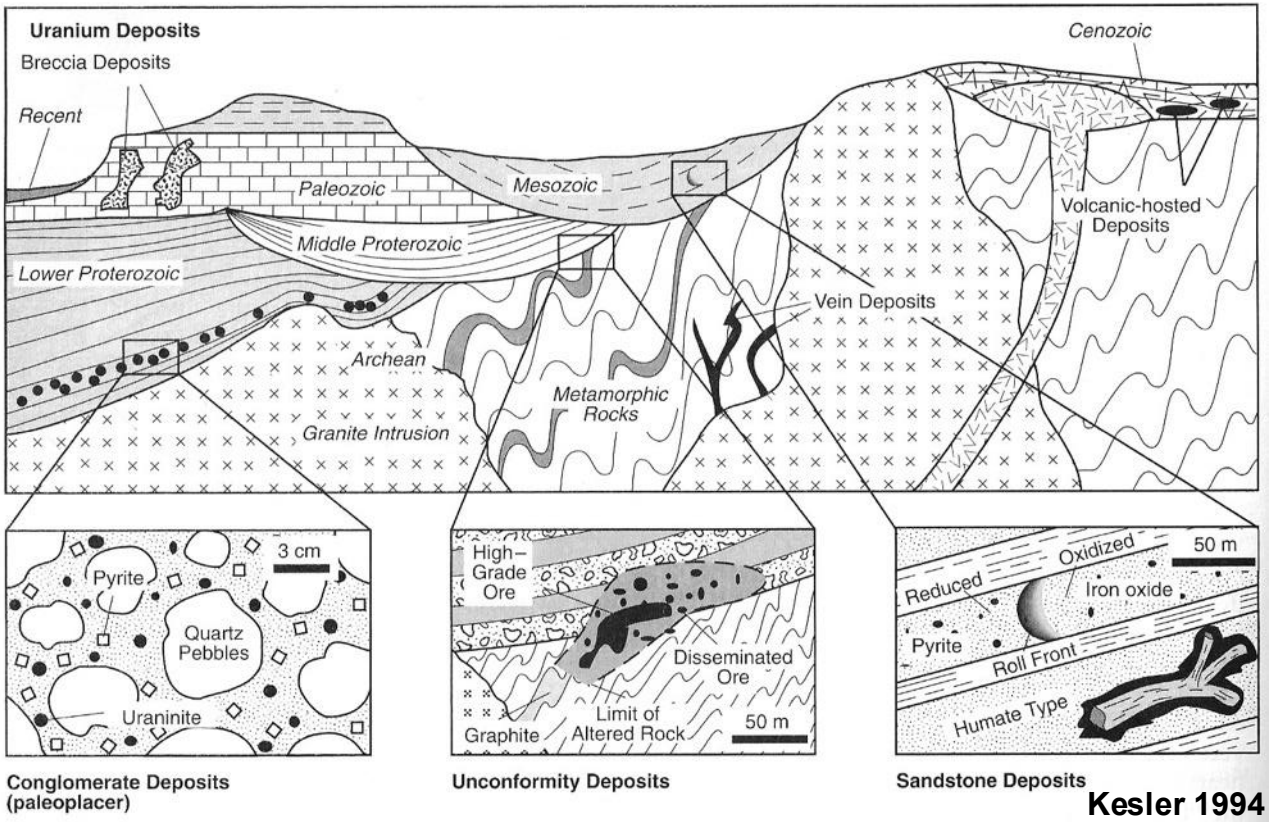
## Lecture 7-3: Uranium deposits



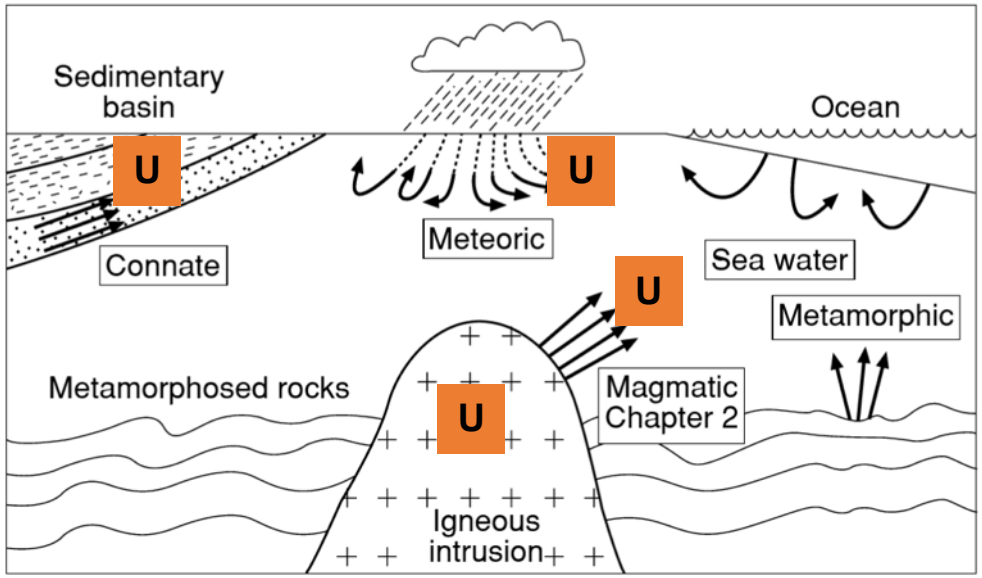
By Ikiwaner - https://commons.wikimedia.org/w/index.php?curid=8132735

# Uranium
## Deposit types

- Magmatic
- Magmatic-hydrothermal
  - Veins associated with igneous rocks
  - Metasomatic deposits
  - Volcanic breccia deposits
- Hydrothermal
  - Unconformity related
  - Roll front/sandstone deposits
- Sedimentary-surface
  - Placers
  - Lignite related
  - Weathering deposits
- Majority of economic deposits are sediment or basement hosted
- Circulation of heated aqueous fluids of variable (surface to igneous) origin – hydrothermal deposits
- Redox behavior of U is key factor

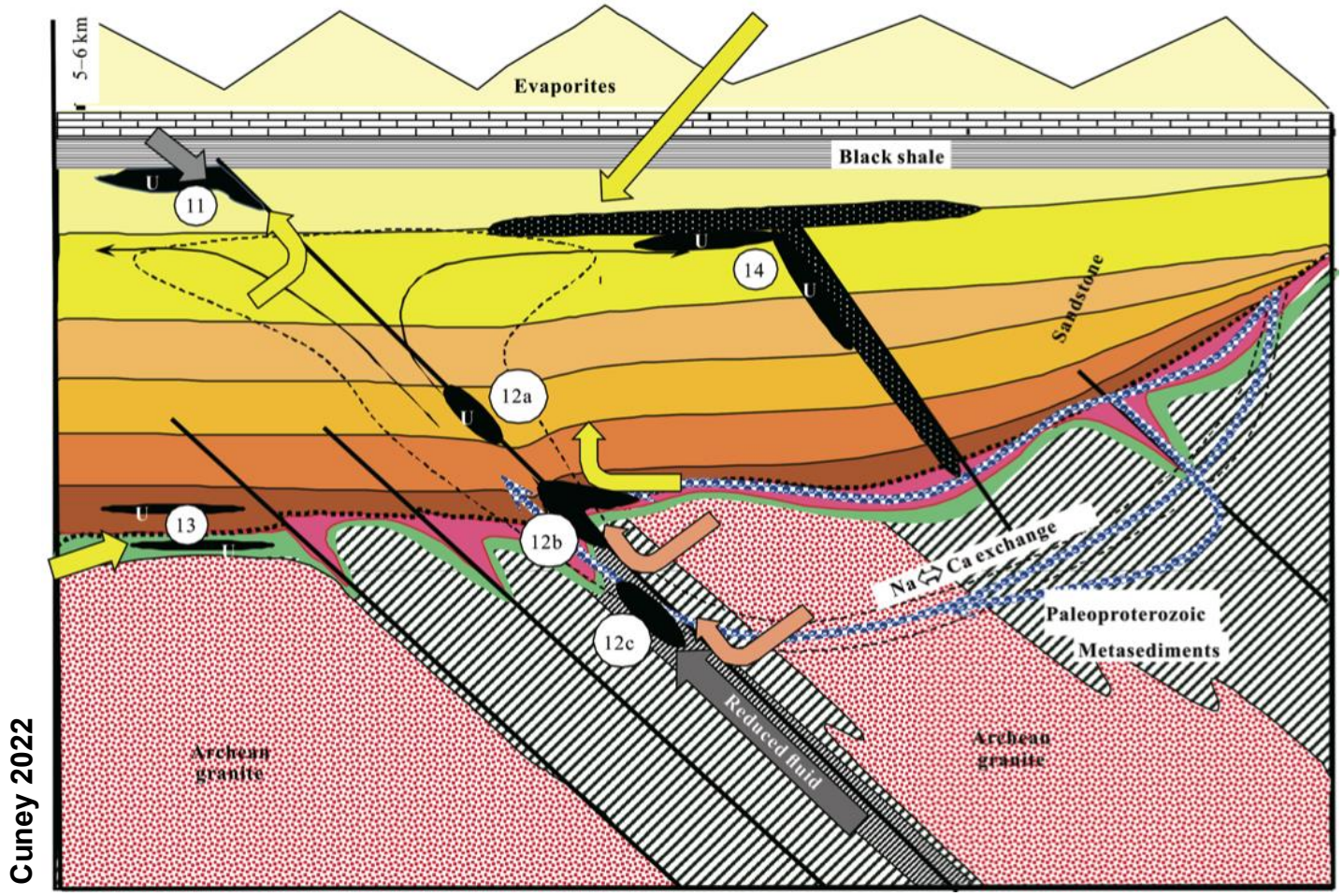



Kesler 1994



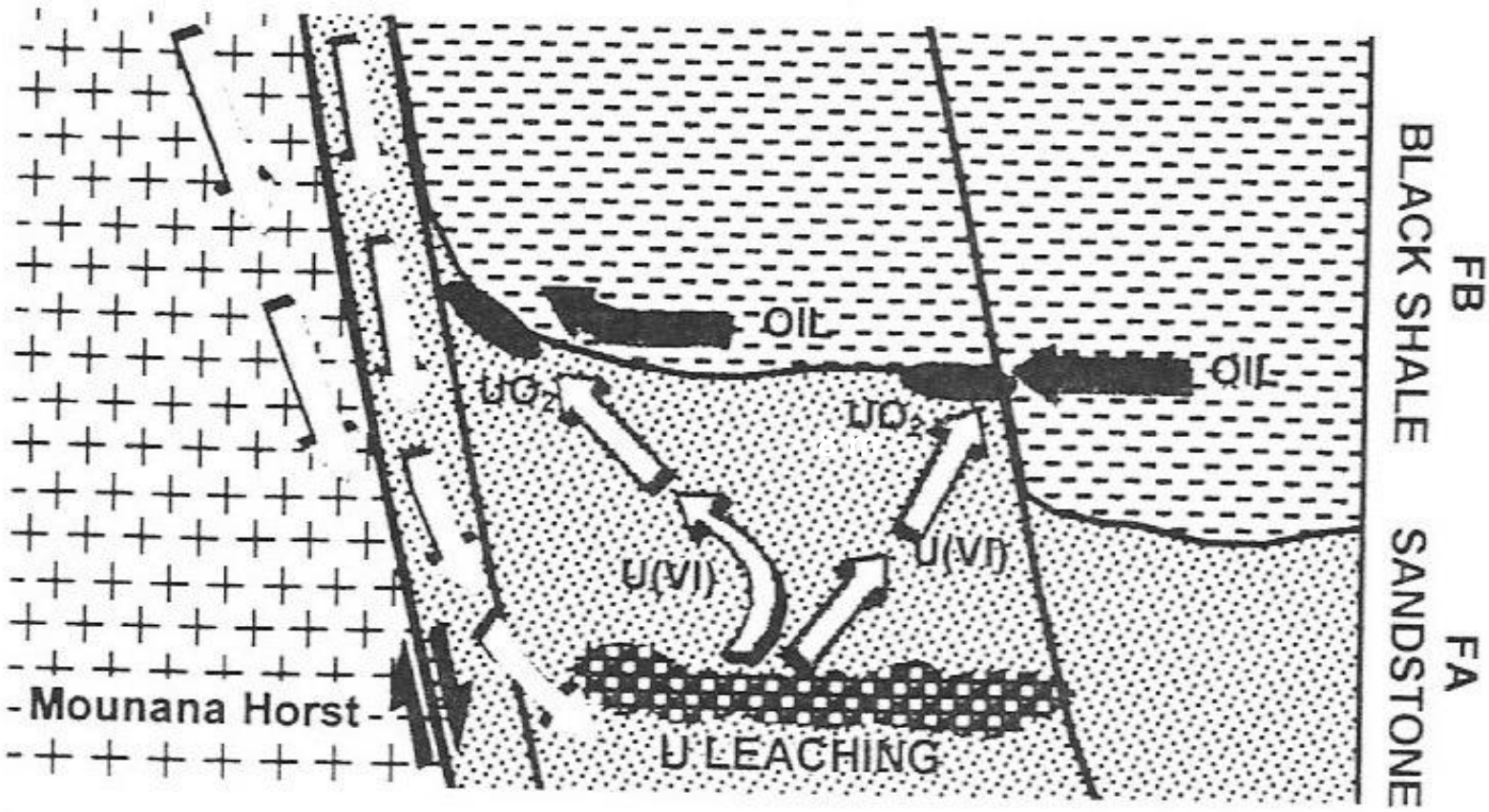

Robb 2005

# Deposit types
## Unconformity deposits



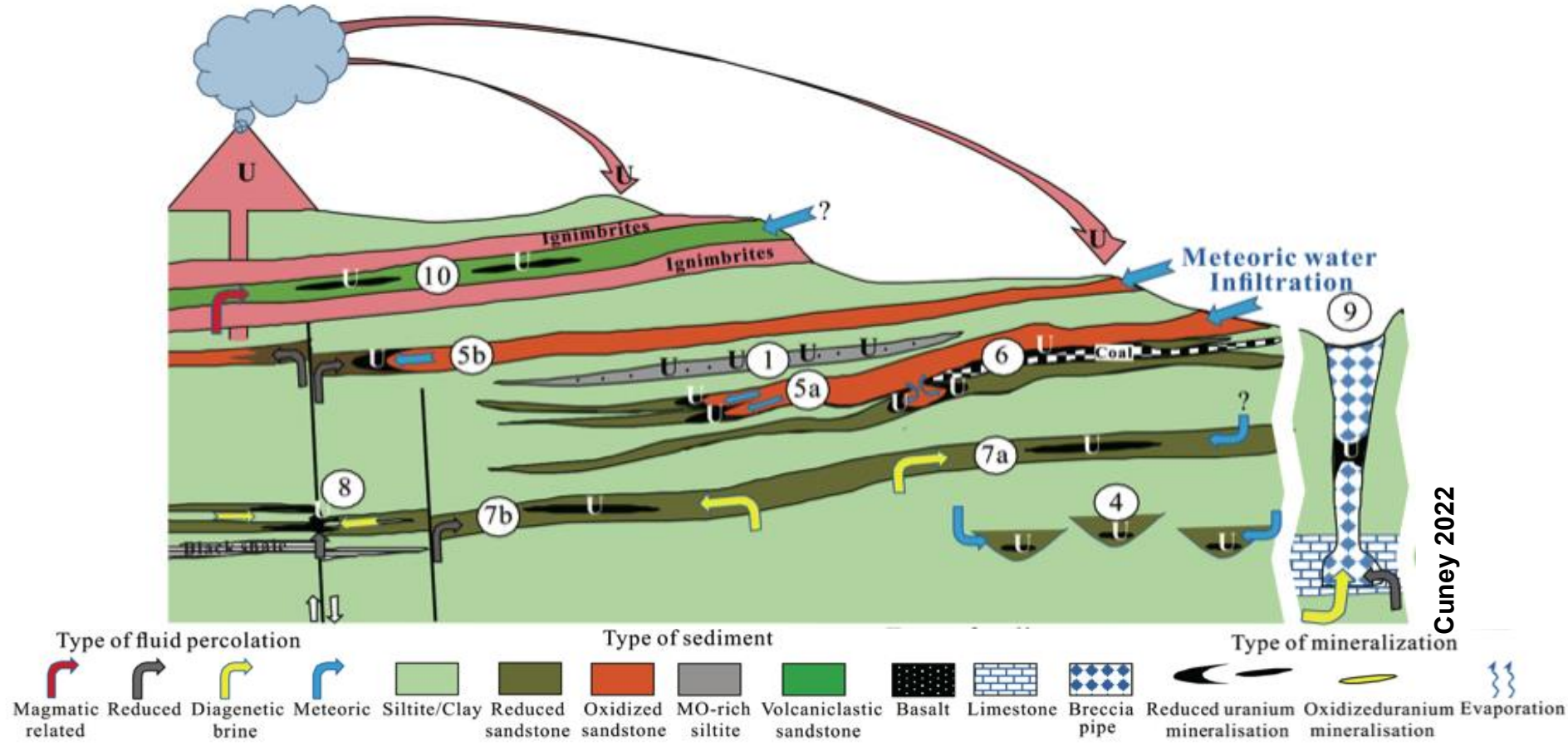

- In basement or above unconformity to Proterozoic sediments; structural control
- Hydrothermal circulation of reducing/oxidizing fluids and precipitation at redox front; epigenetic
- ~33% of world resources (2007); major portion of low-cost U: some of largest and richest U deposits, e.g. Athabasca Basin (Canada)
- Grades 0.1-12% U, associated Ni, Co, As

# Deposit types
## Oklo



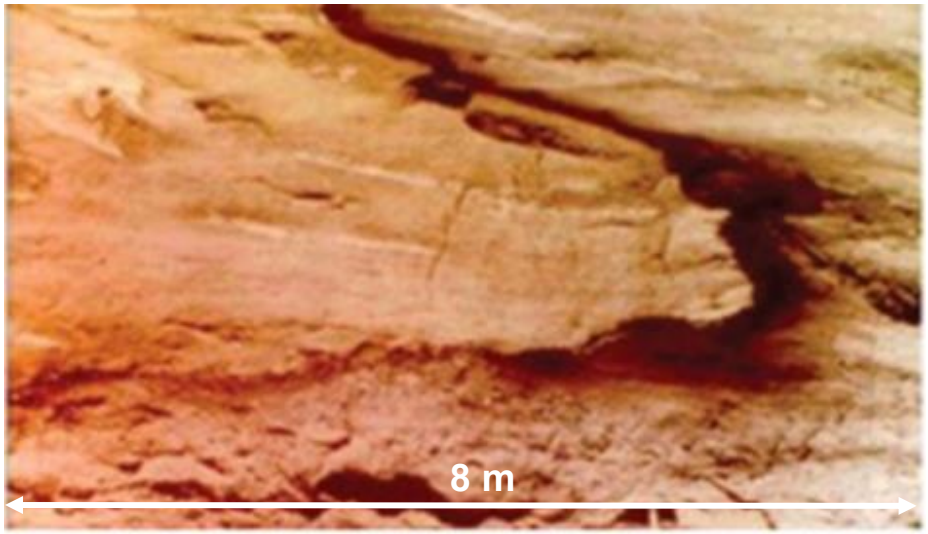

Janeczek 1999

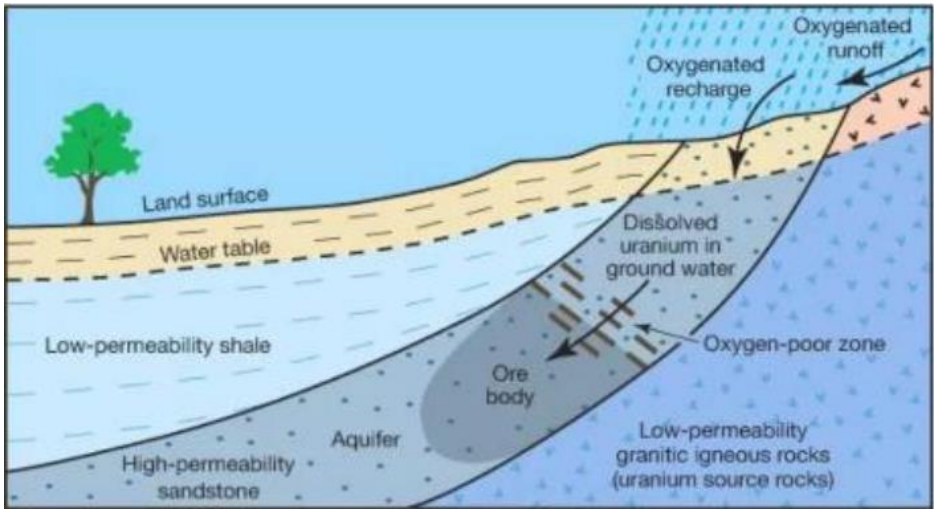
# Deposit types
## Sandstone deposits



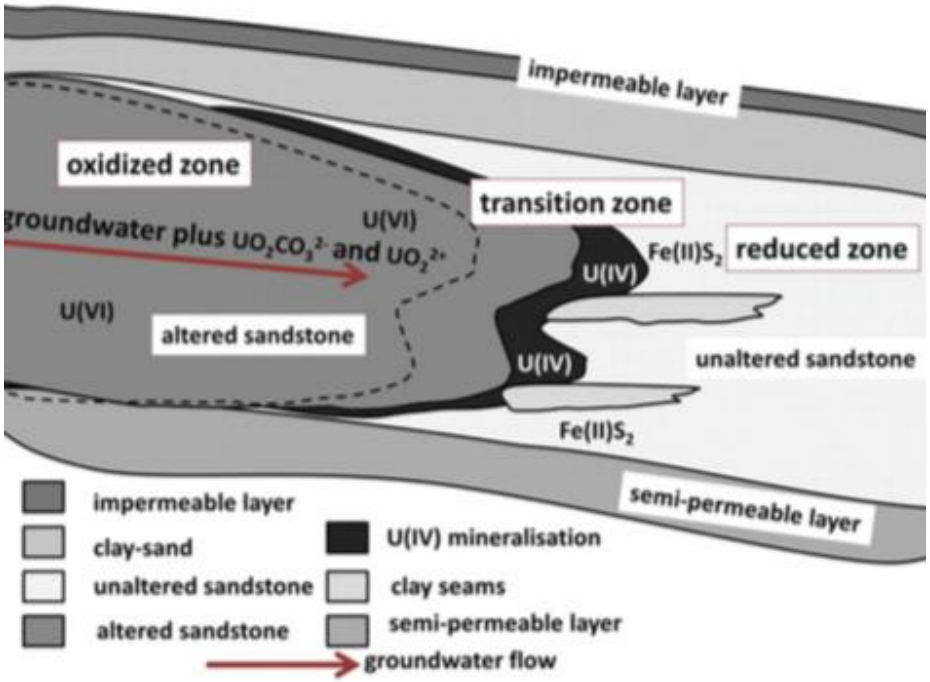

- Circulation of oxidizing meteoric water in sandstone aquifer (U source), precipitation by interaction with reductant (organic matter, hydrocarbons, methane)
- Commonly associated with deeper aqueous fluids (reducing)
- Epigenetic to syngenetic
- Most common: roll front and tabular deposits

# Deposit types
## Roll front deposits



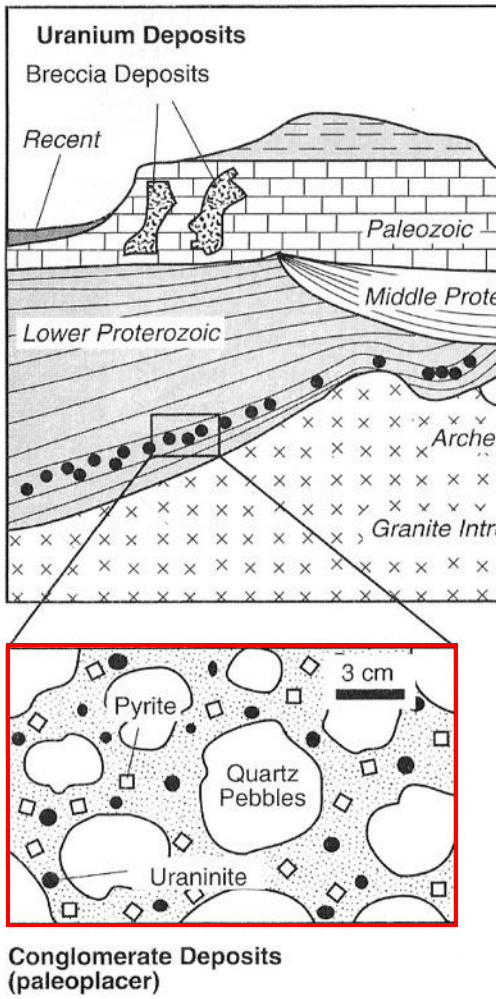




Source: Curnamona Energy

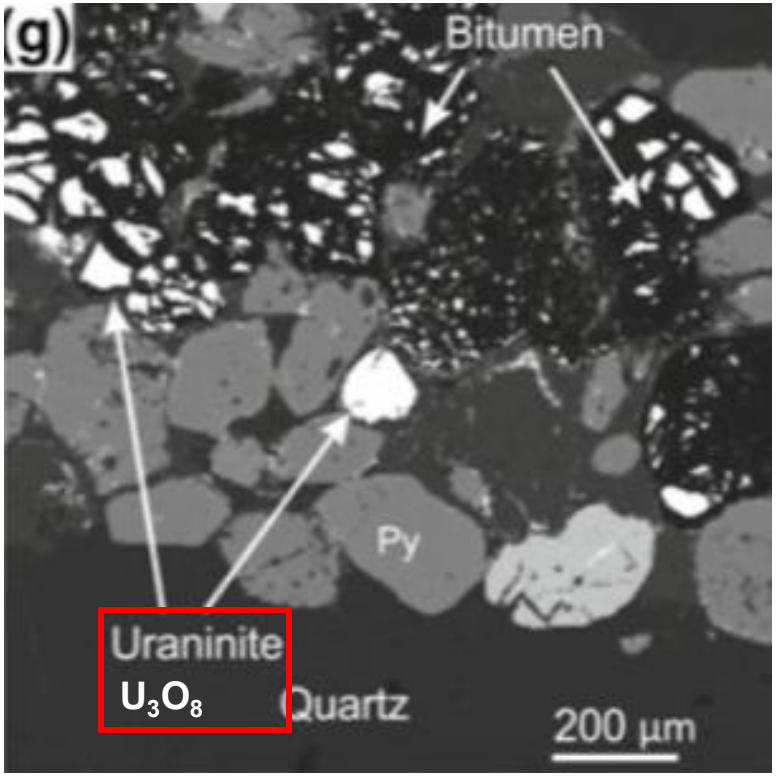



- Circulation of oxidizing groundwater in sandstone aquifer
- U from granites
- Precipitation at curved redox front
- Also associated with decaying organic detritus
- Microbial reactions?
- ~18% of world resources (2007)
- e.g. Colorado Plateau, USA, Kazakhstan
- Grades 0.05-0.4% U

# Deposit types
## Placer



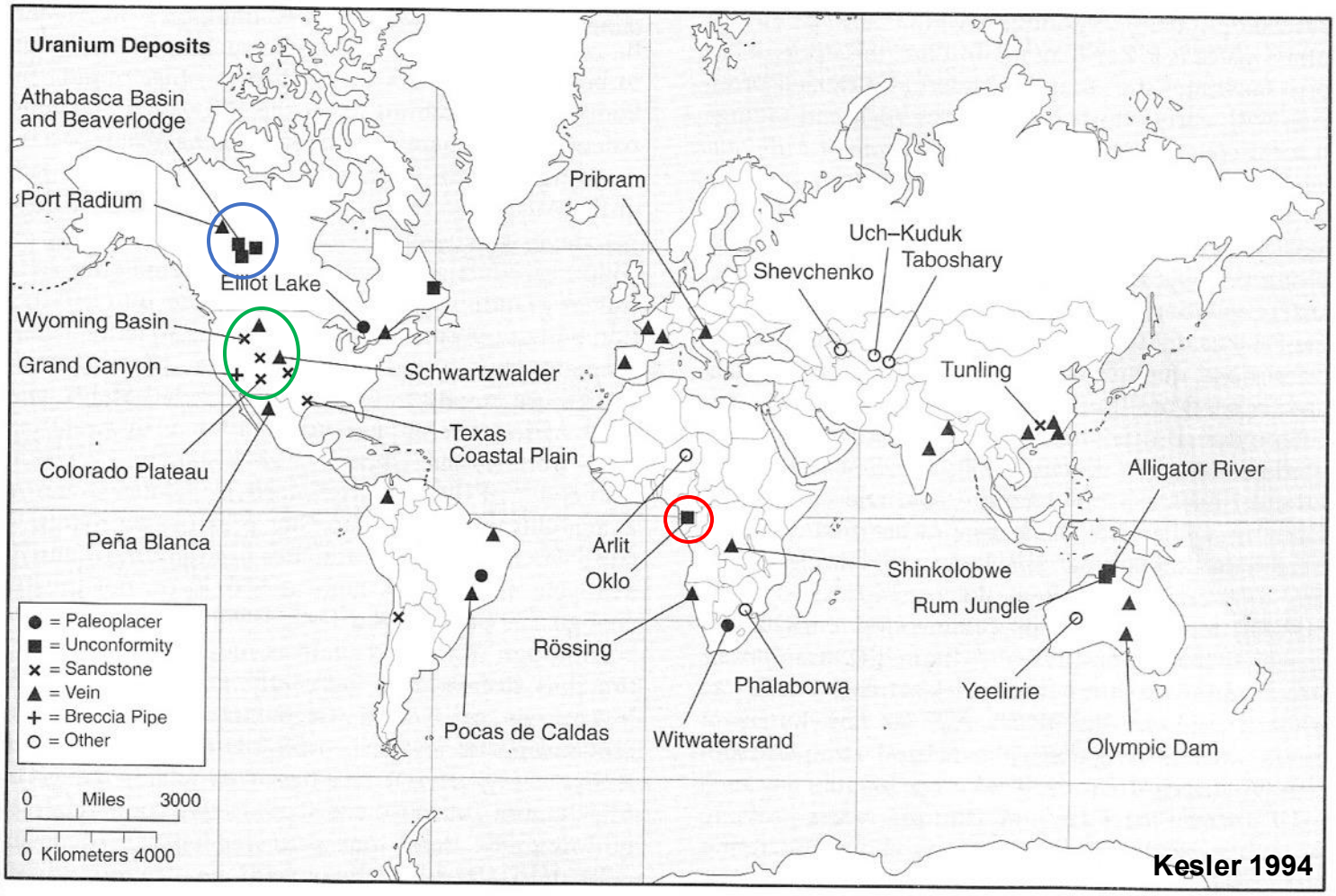

Conglomerate Deposits (paleoplacer)




Frimmel 2019

- U associated with quartz pebbles, occurs mostly as uraninite ($U_3O_8$)
- ~13% of world resources (2007)
- Grades:
  - ~0.01% $U_3O_8$) as by-product
  - ~0.35% $U_3O_8$) where mined exclusively for U
- e.g. Witwatersrand South Africa, Eliot Lake Canada

# Uranium
## Deposit types




- **Wide variety of deposit types; majority linked to shallow crustal processes: access for oxidizing surface waters – redox processes**
- **Sedimentary, weathering, hydrothermal processes, and related to igneous processes**
- **Temporal change from detrital placer processes (reducing Earth) to hydrothermal and sedimentary processes (oxidizing Earth)**

# EART 26001
# Energy and Resources

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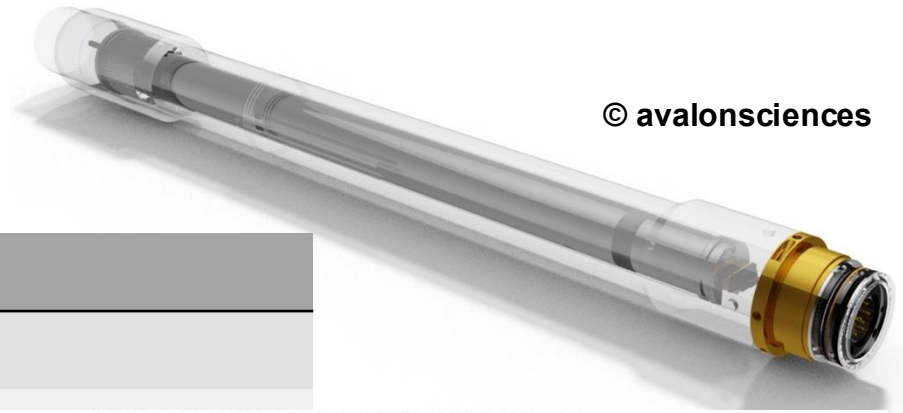
## Lecture 7-4: Uranium exploration, production and environmental aspects



By Ikiwaner - https://commons.wikimedia.org/w/index.php?curid=8132735

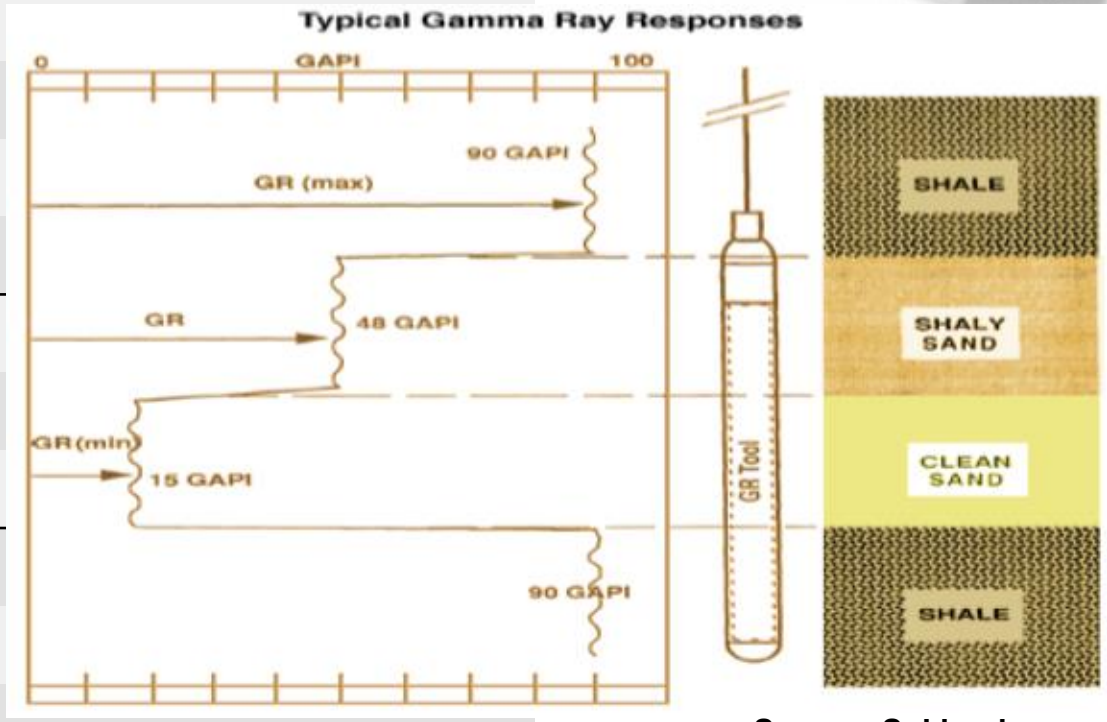
# Exploration
## Geophysical techniques – radiation spectrometry

- **measures natural radioactivity of rocks and formations, emitted mainly by elements K, Th, U (gamma-ray)**



© avalonsciences

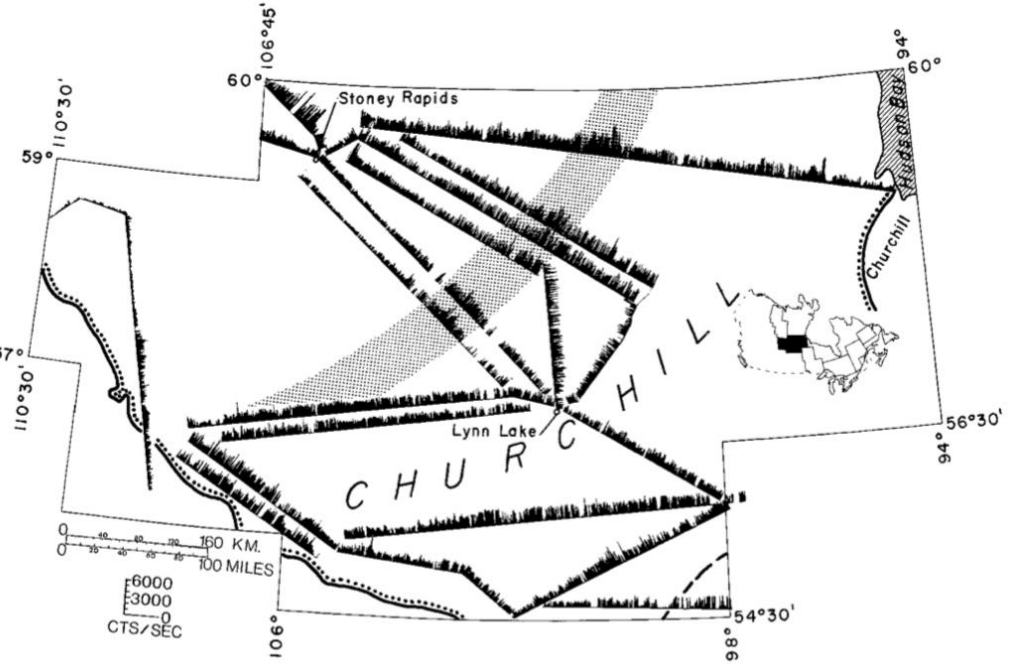
| Element | Source |
|---|---|
| | Shales (illites) |
| | K-feldspars |
| **K** | K-evaporites |
| | Micas |
| | Other K-minerals |
| | Detrital shales |
| **Th** | Th minerals |
| | heavy minerals |
| | Organic shales |
| **U** | Organic matter |
| | U minerals |




Source: Schlumberger

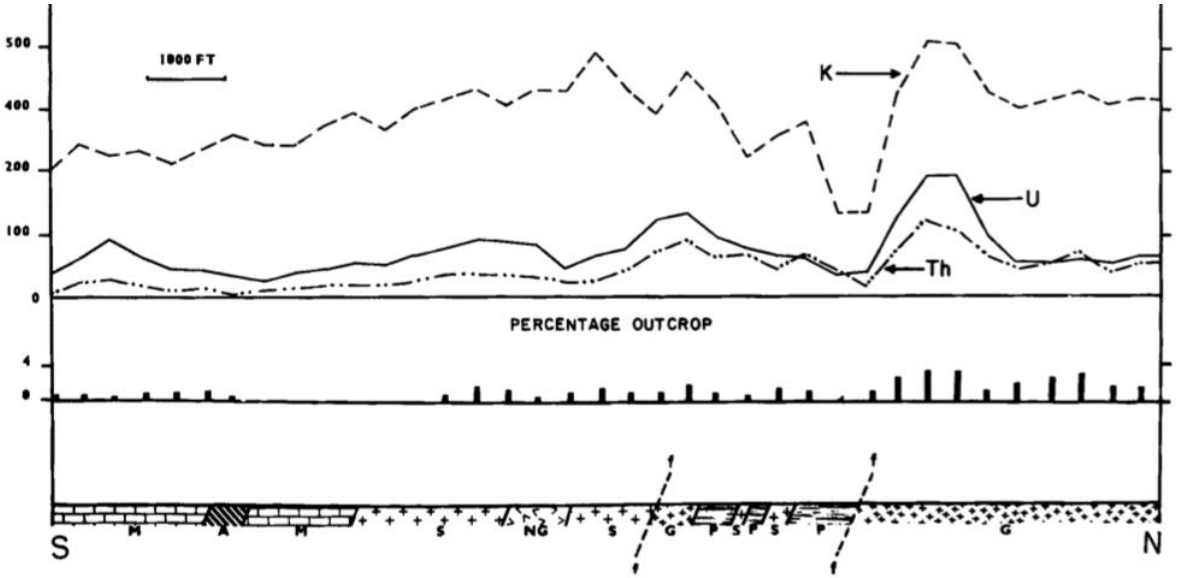
# Exploration
## Geophysical techniques – radiation spectrometry

- Used in airborne radiation surveys




**Typical exploration will involve:**

- Geological mapping
- Airborne radiometric, gravimetric, magnetic, resistivity surveys
- Geochemical mapping (U in streams, soil gas radon, He)
- Drilling




Killeen 1979

# Production
## Mining

- Low grades: volume- and energy-intensive operations
- 2010 global supply figures:
  - 60% open pit and underground mining
  - 36% leaching methods (in situ leaching - ISL)

- Open pit:
  - Ore body near surface
  - Blasting and excavating
  - Water spraying to limit airborne radioactive dust

- Underground:
  - Deeper ore body
  - Main hazard is radon gas: ventilation

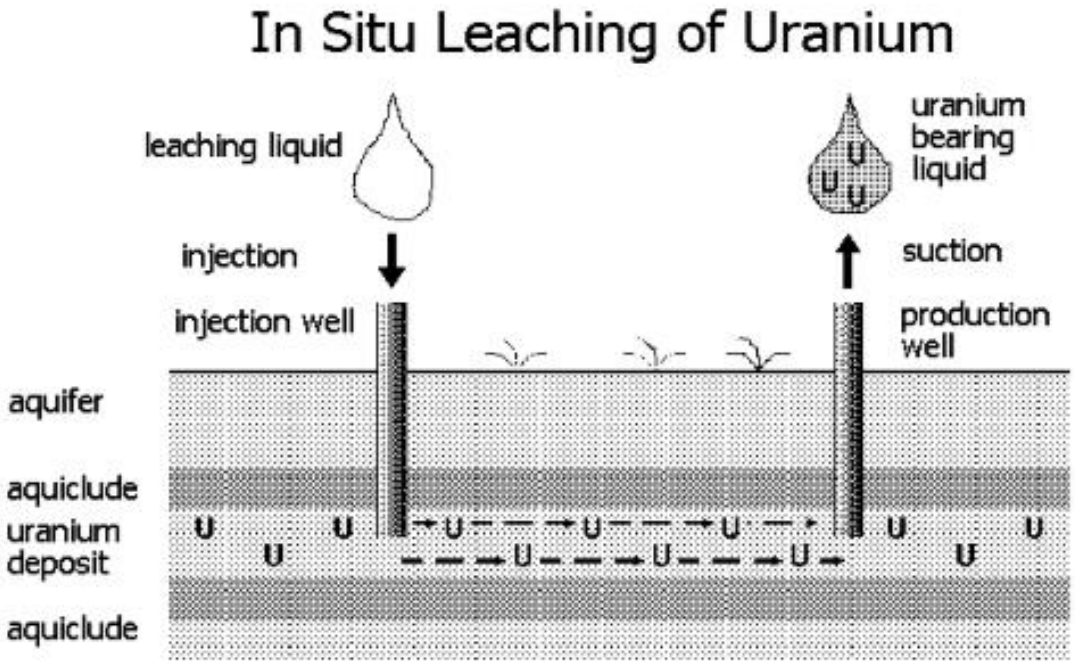


# Production
## Mining

- Low grades: volume- and energy-intensive operations
- 2010 global supply figures:
  - 60% open pit and underground mining
  - 36% leaching methods

- Heap leaching:
  - Acid sprayed onto heap
  - Not used in industrial mining, but small-scale artisanal mining

- In situ leaching (ISL):
  - Acidic or alkaline, $O_2$-enriched water pumped into deposit and recovered (soda, $CO_2$, sulfuric acid)
  - Applicable in permeable deposits
  - Cost-effective, low surface environmental impact; groundwater?




*Principle scheme for an in-situ leaching operation*

Höök 2007

# Production
## Beneficiation

- Processing on site to yellowcake: ammonium diuranate (70-90% $U_3O_8$)
- Enriched in country of use to 2-4% $^{235}U$: gas diffusion or centrifuges
- Significant infrastructure and energy needed for enrichment





# Environmental aspects
## Uranium in environment

- Inhalation of radioactive aerosols during mining: carcinogen, respiratory diseases
- Radiation hazard from tailings: ~20% of radiation are removed in beneficiation, powdered form is easier to get airborne
- Toxic in groundwater
- Strict environmental regulations for waste tailings and working conditions (ventilation, sealing of tailings…)



# Uranium deposits
## Key learning points week 7

- The dominant use of uranium is as fuel for nuclear power generation.
- Uranium is concentrated by igneous processes associated with granitic rocks (lithophile behavior of U), and by redox processes in sedimentary and hydrothermal ore-forming environments. A variety of uranium deposit types exist as a consequence.
- Natural reactors developed criticality and nuclear chain reaction in Gabon ~2.0 Ga ago. Key factors to reach criticality were:
    - U enrichment factor
    - relatively high $^{235}U/^{238}U$ ratio
    - minimum ore thickness
    - high water/rock ratio
- The study of natural reactors provides insights into the geological processes for uranium enrichment, physics of the chain reaction, and conditions for nuclear waste storage.
- Uranium is radioactive and toxic, which has major environmental implications from mining through to radioactive waste disposal.